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Indira Gandhi National Open University
School of Computer and Information
Sciences (SOCIS)

MCS-221
Data Warehousing and Data Mining

DATA WAREHOUSE FUNDAMENTALS AND ARCHITECTURE

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Block

1

DATA WAREHOUSE FUNDAMENTALS AND ARCHITECTURE

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COURSE INTRODUCTION

This course is of 4 credits, divided into two parts – first part (2 credits) covering the Data Warehousing and second part (2 credits) covering the Data Mining.

A data warehouse is a system that stores data from a company's operational databases as well as external sources. Data warehouse platforms are different from operational databases because they store historical information, making it easier for business leaders to analyze data over a specific period of time. Data warehouse platforms also sort data based on different subject matter, such as customers, products or business activities.

Many global corporations have turned to data warehousing to organize data that streams in from corporate branches and operations centers around the world. It's essential for IT students to understand how data warehousing helps businesses remain competitive in a quickly evolving global marketplace. Data warehousing is an increasingly important business intelligence tool which enables historical insights, ensure consistency, allow organizations to make better business decisions, decrease costs, maximize efficiency, increase the power and speed of data analytics, provides major competitive edge and increase sales to improve the bottom line.

It is necessary to choose adequate Data Mining algorithms for making Data Warehouse more useful. Data mining algorithms are used for transforming data into business information and thereby improving decision making process. Data Mining is a set of methods used for data analysis, created with the aim to find out specific dependence, relations and rules related to data and making them out in the new higher level quality information. Data Mining gives results that show the interdependence and relations of data. These dependences are mainly based on various mathematical and statistical relations. Data are collected from internal database and converted into various documents, reports, list etc. which can be further used in decision making processes. After selecting the data for analysis, Data Mining is applied to the appropriate rules of behavior and patterns. That is the reasons why Data Mining is also known as extraction of knowledge, data archeology or pattern analysis. Data mining helps to develop smart market decision, run accurate campaigns, make predictions, and more. With the help of Data mining, we can analyze customer behaviors and their insights. This leads to great success and data-driven business.

The course is organized into 4 Blocks:

Block 1 covers the Introductory topics on Data Warehousing, Data Warehouse architecture, Data Marts and Dimensional Modeling.

Block 2 covers the Extract, Transform and Loading (ETL) aspects of Data Warehousing, Online Analytical Processing and some Trends in Data Warehouse.

Block 3 covers the introductory topics related to Data Mining, Data Preprocessing and Mining Frequent Patterns and Associations

Block 4 covers the Classification, Clustering of Data Mining, Text and Web Mining.

There is a lab component associated with this course (i.e., Section-2 Data Mining Lab of MCSL-223 course).

BLOCK INTRODUCTION

The title of the Block is Data Warehouse Fundamentals and Architecture. The objectives of this block are to make you understand about the underlying concepts of Data Warehousing, identify the components of the Data Warehouse Architecture, to know the difference between the Data Warehouse and Data Marts, to understand the Data Warehouse Development Life Cycle and to elucidate the dimensional modeling techniques.

The block is organized into 3 units:

Unit 1 covers the fundamentals of data warehousing, its evolution, characteristics of data warehousing, online transaction processing systems and applications of data warehouses;

Unit 2 covers the data warehouse architecture, data marts and data warehouse development life cycle; and

Unit 3 covers the introduction to dimensional modeling, identifying facts and dimensions, star schema, snowflake schema and fact constellation schema.

UNIT 1 FUNDAMENTALS OF DATA WAREHOUSE

Structure

1.0 INTRODUCTION

A database often contains information or data collection that is generally stored electronically in a computer system. It is easy to access, manage, modify, update, monitor, and organize the data. Data is stored in the tables of the database.

The process of consolidating data and analyzing it to obtain some insights has been around for centuries, but we just recently began referring to this as data warehousing. Any operational or transactional system is only designed with its own functionality and hence, it could handle limited amounts of data for a limited amount of time. The operational systems are not designed or architected for long term data retention as the historical data is little to no importance to them. However, to gain a point-in-time visibility and understand the high-level operational aspects of any business, the historical data plays a vital role. With the emergence of matured Relational Database Management Systems (RDBMS) in 1960s, engineers across various enterprises started architecting ways to copy the data from the transactional systems over to different databases via manual or automated mechanism and use it for reporting and analysis. As the data in the transactional systems would get purged periodically, it would not be the case in these analytical repositories as their purpose was to store as much data as possible; hence the word "data warehouse" came into existence because these repositories would become a warehouse for the data.

Data Warehousing (DW) as a practice became very prominent during late 80s when the enterprises started building decision support systems that were mainly responsible to support reporting. As there was a rapid advancement in the performance of these relational database during late 1990s and early 2000s, Data Warehousing became a core part of the Information Technology group across large enterprises. In fact, some of the vendors like Netezza, Teradata started offering customized hardware to manage data warehouse architectures within state-of-the-art machines. Data Warehousing had evolved to be on top of the list of priorities since mid 2000s. Data supply chain ecosystem has grown exponentially in the current world and so is the way enterprises architect their data warehouses.

A well architected data warehouse serves as an extended vision for the enterprise where multiple departments can gain actionable insights to manage key business decisions that could drive operational excellence or revenue generating opportunities for the enterprise.

This unit covers the basic features of data warehousing, its evolution, characteristics, online transaction processing (OLTP), online analytical processing, popular platforms and applications of data warehouses.

1.1 OBJECTIVES

After going through this unit, you shall be able to:

- understand the evolution of data warehouse;
- describe various characteristics of data warehouse;
- list the benefits and applications of a data warehouse;
- discuss the significance of metadata in data ware house;
- list and discuss the types of data warehouses, and
- identify the popular data warehouse platforms;

1.2 EVOLUTION OF DATA WAREHOUSE

The relational database revolution in the early 1980s ushered in an era of improved access to the valuable information contained deep within data. It was soon discovered that databases modeled to be efficient at transactional processing were not always optimized for complex reporting or analytical needs.

In fact, the need for systems offering decision support functionality predates the first relational model and SQL. But the practice known today as Data Warehousing really saw its genesis in the late 1980s. An IBM Systems Journal article published in 1988, An architecture for a business information system coined the term "business data warehouse," although a future progenitor of the practice, Bill Inmon, used a similar term in the 1970s. Considered by many to be the Father of Data Warehousing, Bill Inmon, an American Computer Scientist is first began to discuss the principles around the Data Warehouse and even coined the term. Throughout the latter 1970s into the 1980s, Inmon worked extensively as a data professional, honing his expertise in all manners of relational Data Modeling. Inmon's work as a Data Warehousing pioneer took off in the early 1990s when he ventured out on his own, forming his first company, Prism Solutions. One of Prism's main products was the Prism Warehouse Manager, one of the first industry tools for creating and managing a Data Warehouse.

In 1992, Inmon published Building the Data Warehouse, one of the seminal volumes of the industry. Later in the 1990s, Inmon developed the concept of the Corporate Information Factory, an enterprise level view of an organization's data of which Data Warehousing plays one part. Inmon's approach to Data Warehouse design focuses on a centralized data repository modeled to the third normal form. Inmon's approach is often characterized as a top-down approach. Inmon feels using strong relational modeling leads to enterprise-wide consistency facilitating easier development of individual data marts to better serve the needs of the departments using the actual data. This approach differs in some respects to the "other" father of Data Warehousing, Ralph Kimball.

While Inmon's Building the Data Warehouse provided a robust theoretical background for the concepts surrounding Data Warehousing, it was Ralph Kimball's The Data Warehouse Toolkit, first published in 1996, that included a host of industry-honed, practical examples for OLAP-style modeling. Kimball, on the other hand, favors the development of individual data marts at the departmental level that get integrated together using the Information Bus architecture. This bottom up approach fits-in nicely with Kimball's preference for star-schema modeling. Both approaches remain core to Data Warehousing architecture as it stands today. Smaller firms might find Kimball's data mart approach to be easier to implement with a constrained budget. Dimensional modeling in many cases is easier for the end user to understand.

According to Bill Inmon, "A warehouse is a subject-oriented, integrated, time-variant and non-volatile collection of data in support of management's decision making process".

According to Ralph Kimball, "Data Warehouse (DW) is the conglomerate of all data marts within the enterprise. Information is always stored in the dimensional model".

1.3 DATA WAREHOUSING AND ITS NEED

Data Warehouse is used to collect and manage data from various sources, in order to provide meaningful business insights. A data warehouse is usually used for linking and analyzing heterogeneous sources of business data. The data warehouse is the center of the data collection and reporting framework developed for the BI system. Data warehouse systems are real-time repositories of information, which are likely to be tied to specific applications. Data warehouses gather data from multiple sources (including databases), with an emphasis on storing, filtering, retrieving and in particular, analyzing huge quantities of organized data. The data warehouse operates in information-rich environment that provides an overview of the company, makes the current and historical data of the company available for decisions, enables decision support transactions without obstructing operating systems, makes information consistent for the organization, and presents a flexible and interactive information source.

1.3.1 Need for Data Warehouse

Data warehouses are used extensively in the largest and most complex businesses around the world. In demanding situations, good decision making becomes critical. Significant and relevant data is required to make decisions. This is possible only with the help of a well-designed data warehouse. Following are some of the reasons for the need of Data Warehouses:

Enhancing the turnaround time for analysis and reporting: Data warehouse allows business users to access critical data from a single source enabling them to take quick decisions. They need not waste time retrieving data from multiple sources. The business executives can query the data themselves with minimal or no support from IT which in turn saves money and time.

Improved Business Intelligence: Data warehouse helps in achieving the vision for the managers and business executives. Outcomes that affect the strategy and procedures of an organization will be based on reliable facts and supported with evidence and organizational data.

Benefit of historical data: Transactional data stores data on a day to day basis or for a very short period of duration without the inclusion of historical data. In comparison, a data warehouse stores large amounts of historical data which enables the business to include time-period analysis, trend analysis, and trend forecasts.

Standardization of data: The data from heterogeneous sources are available in a single format in a data warehouse. This simplifies the readability and accessibility of data. For example, gender is denoted as Male/ Female in Source 1 and m/f in Source 2 but in a data warehouse the gender is stored in a format which is common across all the businesses i.e. M/F.

Immense ROI (Return On Investment): Return On Investment refers to the additional revenues or reduces expenses a business will be able to realize from any project.

Now, let us study the benefits.

1.3.2 Benefits of Data Warehouse

Several enterprises adopt data warehousing as it offers many benefits, such as streamlining the business and increasing profits. Following are some of the benefits of having a data warehouse:

Scalability - Businesses today cannot survive for long if they cannot easily expand and scale to match the increase in the volume of daily transactions. DW is easy to scale, making it easier for the business to stride ahead with minimum hassle.

Access to Historical Insights - Though real-time data is important, historical insights cannot be ignored when tracing patterns. Data warehousing allows businesses to access past data with just a few clicks. Data that are months and years old can be stored in the warehouse.

Works On-Premises and on Cloud - Data warehouses can be built on-premises or on cloud platforms. Enterprises can choose either option, depending on their existing business system and the long-term plan. Some businesses rely on both.

Better Efficiency - Data warehousing increases the efficiency of the business by collecting data from multiple sources and processing it to provide reliable and actionable insights. The top management uses these insights to make better and faster decisions, resulting in more productivity and improved performance.

Improved Data Security - Data security is crucial in every enterprise. By collecting data in a centralized warehouse, it becomes easier to set up a multi-level security system to prevent the data from being misused. Provide restricted access to data based on the roles and responsibilities of the employees.

Increase Revenue and Returns - When the management and employees have access to valuable data analytics, their decisions and actions will strengthen the business. This increases the revenue in the long run.

Faster and Accurate Data Analytics - When data is available in the central data warehouse, it takes less time to perform data analysis and generate reports. Since the data is already cleaned and formatted, the results will be more accurate.

Let us study the various approaches in detail in the following section.

1.4 DATA WAREHOUSE DESIGN APPROACHES

Data Warehouse design approaches are very important aspect of building data warehouse. Selection of right data warehouse design could save lot of time and project cost.

There are two different Data Warehouse Design Approaches normally followed when designing a Data Warehouse solution and based on the requirements of your project you can choose which one suits your particular scenario. These methodologies are a result of research from Bill Inmon (Top-Down Approach) and Ralph Kimball(Bottom up Approach).

1.4.1 Top-down Approach

Bill Inmon's design methodology is based on a top-down approach which is illustrated in the Figure 1. In the top-down approach, the data warehouse is designed first and then data mart is built on top of data warehouse.

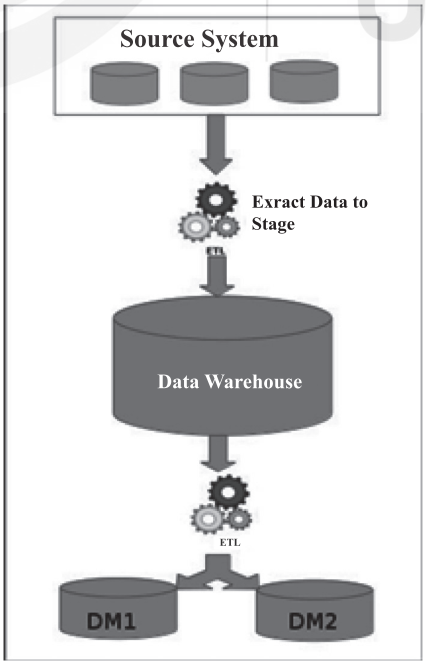

Figure 1: Top-Down DW Design Approach

Below are the steps that are involved in top-down approach:

- Data is extracted from the various source systems. The extracts are loaded and validated in the stage area. Validation is required to make sure the extracted data is accurate and correct. You can use the ETL tools or approach to extract and push to the data warehouse.
- Data is extracted from the data warehouse in regular basis in stage area. At this step, you will apply various aggregation, summarization techniques on extracted data and loaded back to the data warehouse.
- Once the aggregation and summarization is completed, various data marts extract that data and apply the some more transformation to make the data structure as defined by the data marts.

1.4.2 Bottom-up Approach

Ralph Kimball's data warehouse design approach is called dimensional modelling or the Kimball methodology which is illustrated in Figure 2. This methodology follows the bottom-up approach.

As per this method, data marts are first created to provide the reporting and analytics capability for specific business process, later with these data marts enterprise data warehouse is created.

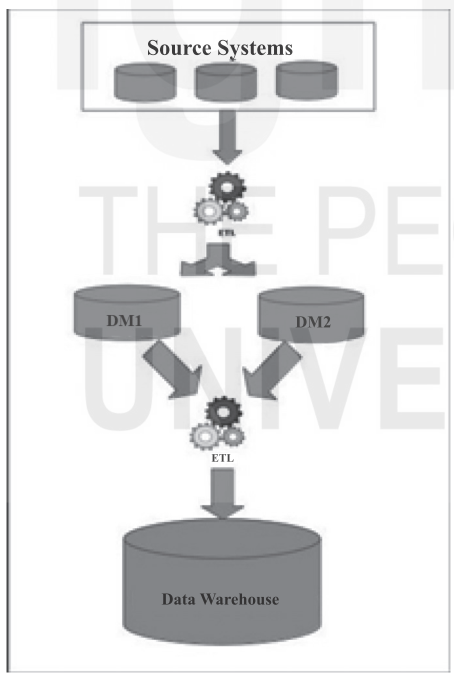

Figure 2: Bottom-Up DW Design Approach

Basically, Kimball model reverses the Inmon model i.e. Data marts are directly loaded with the data from the source systems and then ETL process is used to load in to Data Warehouse. The above image depicts how the top-down approach works.

Below are the steps that are involved in bottom-up approach:

- The data flow in the bottom up approach starts from extraction of data from various source systems into the stage area where it is processed and loaded into the data marts that are handling specific business process.
- After data marts are refreshed the current data is once again extracted in stage area and transformations are applied to create data into the data mart

structure. The data is the extracted from Data Mart to the staging area is aggregated, summarized and so on loaded into EDW and then made available for the end user for analysis and enables critical business decisions.

- Having discussed the data warehouse design strategies, let us study the characteristics of the DW in the next section.

1.5 CHARACTERISTICS OF A DATA WAREHOUSE

Data warehouses are systems that are concerned with studying, analyzing and presenting enterprise data in a way that enables senior management to make decisions. The data warehouses have four essential characteristics that distinguish them from any other data and these characteristics are as follows:

- **Subject-oriented**

A DW is always a subject-oriented one, as it provides information about a specific theme instead of current organizational operations. On specific themes, it can be done. That means that it is proposed to handle the data warehousing process with a specific theme (subject) that is more defined. Figure 3 shows Sales, Products, Customers and Account are the different themes.

A data warehouse never emphasizes only existing activities. Instead, it focuses on data demonstration and analysis to make different decisions. It also provides an easy and accurate demonstration of specific themes by eliminating information that is not needed to make decisions.

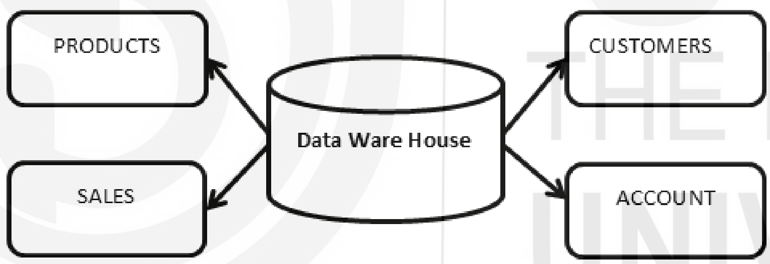

Figure 3: Subject-Oriented Characteristic Feature of a DW

- **Integrated**

Integration involves setting up a common system to measure all similar data from multiple systems. Data was to be shared within several database repositories and must be stored in a secured manner to access by the data warehouse. A data warehouse integrates data from various sources and combines it in a relational database. It must be consistent, readable, and coded.. The data warehouse integrates several subject areas as shown in the figure 4.

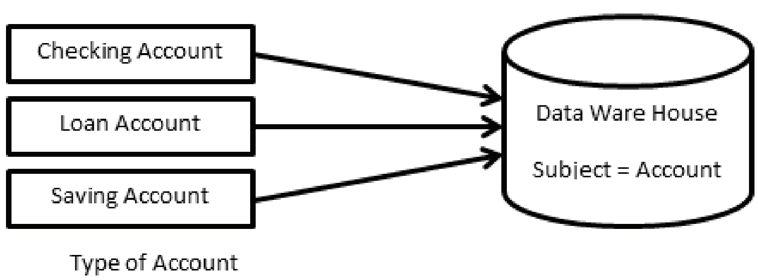

Figure 4: Integrated Characteristic Feature of a DW

- **Time-Variant**

Information may be held in various intervals such as weekly, monthly, and yearly as shown in Figure 5. It provides a series of limited-time, variable rate, online transactions. The data warehouse covers a broader range of data than the operational systems. When the data stored in the data store has a certain amount of time, it can be predictable and provide history. It has aspects of time embedded within it. One other facet of the data warehouse is that the data cannot be changed, modified or updated once it is stored.

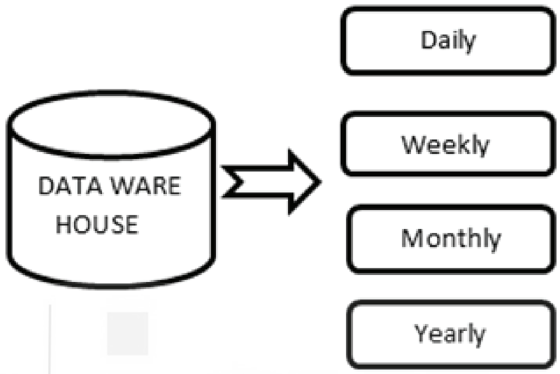

Figure 5: Time- Variant Characteristic Feature of a DW

- **Non-Volatile**

The data residing in the data warehouse is permanent, as the name non -volatile suggests. It also ensures that when new data is added, data is not erased or removed. It requires the mammoth amount of data and analyses the data within the technologies of warehouse. Figure 6 shows the non- volatile data warehouse vs operational database. A data warehouse is kept separate from the operational database and thus the data warehouse does not represent regular changes in the operational database. Data warehouse integration manages different warehouses relevant to the topic.

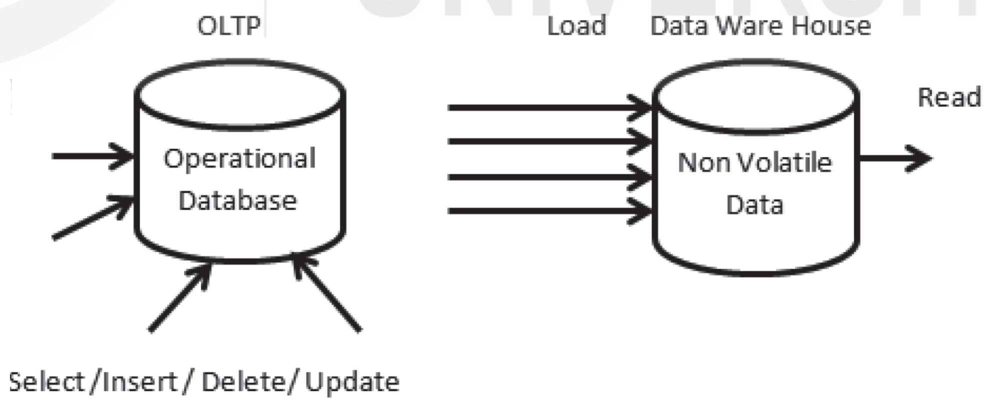

Figure 6: Non –Volatile Characteristic Feature of DW

1.5.1 How Data Warehouse Works?

A data warehouse is a central repository in which one or more sources of information are collected. Data in the data warehouse may be Structured, Semi-structured, or Unstructured. Data are processed, transformed, and accessed by end users for use in business intelligence reporting and decision-making. A data warehouse integrates disparate primary sources into a comprehensive source. Through the integration of all this information, an organization can maintain a more holistic level of customer service. This ensures that all available data is properly considered. Data warehouse enables data mining to find patterns of information that increases profits.

Figure 7 shows the important components of the data warehouse.

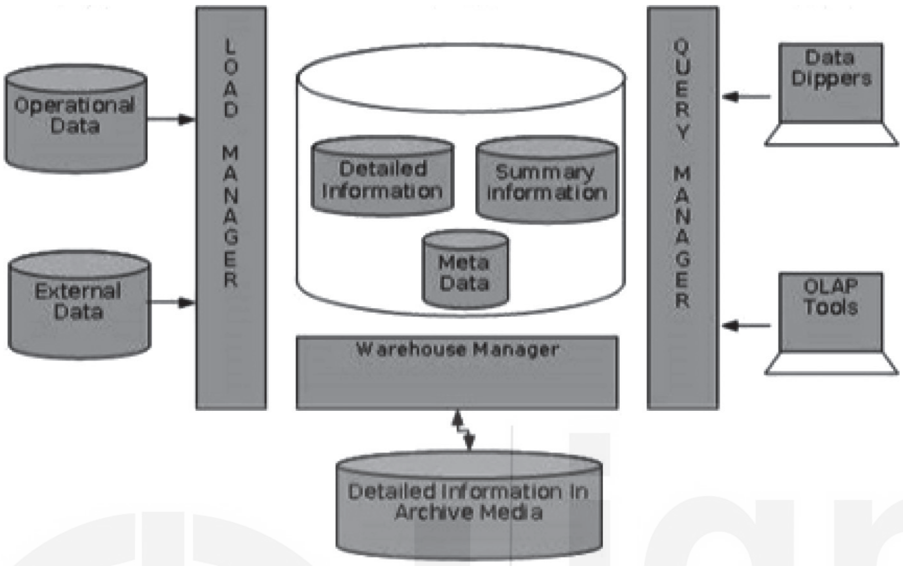

Figure 7: Components of a Data Warehouse

- **Load Manager**

Load Manager Component of data warehouse is responsible for collection of data from operational system and converts them into usable form for the users. This component is responsible for importing and exporting data from operational systems. This component includes all of the programs and applications interfaces that are responsible for pooling the data out of the operational system, preparing it, loading it into warehouse itself it performs the following tasks such as identification of data, validation of data about the accuracy, extraction of data from original source, cleansing of data by eliminating meaningless values and making it usable, data formatting, data standardization by getting them into a consistent form, data merging by taking data from different sources and consolidating into one place and establishing referential integrity.

- **Warehouse Manager**

The warehouse manager is the center of data-warehousing system and is the data warehouse itself. It is a large, physical database that holds a vast amount of information from a wide variety of sources. The data within the data warehouse is organized such that it becomes easy to find, use and update frequently from its sources.

- **Query Manager**

Query Manager Component provides the end-users with access to the stored warehouse information through the use of specialized end-user tools. Data mining access tools have various categories such as query and reporting, on-line analytical processing (OLAP), statistics, data discovery and graphical and geographical information systems.

- **End-user access tools**

This is divided into the following categories, such as:

- Reporting Data
- Query Tools
- Data Dippers
- Tools for EIS
- Tools for OLAP and tools for data mining.

☞ **Check Your Progress 1**

1) What is a Data Warehouse and why is it important?

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2) Mention the characteristics of a Data Warehouse.

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1.6 OLTP AND OLAP

Online Transaction Processing (OLTP) and Online Analytical Processing (OLAP) are the two terms which look similar but refer to different kinds of systems. Online transaction processing (OLTP) captures, stores, and processes data from transactions in real time. Online analytical processing (OLAP) uses complex queries to analyze aggregated historical data from OLTP systems.

1.6.1 Online Transaction Processing (OLTP)

An OLTP system captures and maintains transaction data in a database. Each transaction involves individual database records made up of multiple fields or columns. Examples include banking and credit card activity or retail checkout scanning.

In OLTP, the emphasis is on fast processing, because OLTP databases are read, written, and updated frequently. If a transaction fails, built-in system logic ensures data integrity.

1.6.2 Online Analytical Processing (OLTP)

OLAP applies complex queries to large amounts of historical data, aggregated from OLTP databases and other sources, for data mining, analytics, and business intelligence projects. In OLAP, the emphasis is on response time to these complex queries. Each query involves one or more columns of data aggregated from many rows.

Examples include year-over-year financial performance or marketing lead generation trends. OLAP databases and data warehouses give analysts and decision-makers the ability to use custom reporting tools to turn data into information. Query failure in OLAP does not interrupt or delay transaction processing for customers, but it can delay or impact the accuracy of business intelligence insights.

OLTP is operational, while OLAP is informational. A glance at the key features of both kinds of processing illustrates their fundamental differences, and how they work together. The table (Table 1) below summarizes differences between OLTP and OLAP.

Table 1: OLTP Vs OLAP

	OLTP	**OLAP**
Characteristics	Handles a large number of small transactions	Handles large volumes of data with complex queries
Query types	Simple standardized queries	Complex queries
Operations	Based on INSERT, UPDATE, DELETE commands	Based on SELECT commands to aggregate data for reporting
Response time	Milliseconds	Seconds, minutes, or hours depending on the amount of data to process
Design	Industry-specific, such as retail, manufacturing, or banking	Subject-specific, such as sales, inventory, or marketing
Source	Transactions	Aggregated data from transactions
Purpose	Control and run essential business operations in real time	Plan, solve problems, support decisions, discover hidden insights
Data updates	Short, fast updates initiated by user	Data periodically refreshed with scheduled, long-running batch jobs
Space requirements	Generally small if historical data is archived	Generally large due to aggregating large datasets
Backup and recovery	Regular backups required to ensure business continuity and meet legal and governance requirements	Lost data can be reloaded from OLTP database as needed in lieu of regular backups
Productivity	Increases productivity of end users	Increases productivity of business managers, data analysts, and executives
Data view	Lists day-to-day business transactions	Multi-dimensional view of enterprise data
User examples	Customer-facing personnel, clerks, online shoppers	Knowledge workers such as data analysts, business analysts, and executives
Database design	Normalized databases for efficiency	Denormalized databases for analysis

OLTP provides an immediate record of current business activity, while OLAP generates and validates insights from that data as it's compiled over time. That historical perspective empowers accurate forecasting, but as with all business intelligence, the insights generated with OLAP are only as good as the data pipeline from which they emanate.

☞ **Check Your Progress 2**

1) Why a data warehouse is separated from Operational Databases?

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2) Mention the key differences between a database and a data warehouse.

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1.7 DATA GRANULARITY

Granularity is one of the main elements in the modeling of DW data. Granularity of data refers to detail levels. Multiple levels of detail may be available depending on the requirements. At least two granular levels exist for many data warehouses. The relation between detailing and granularity is important to understand. It means greater detail of the data (less summary) when you speak of less granularity or fine granularity. Greater granularity means fewer details or gross granularity (greater summarization). The operational data is stored at the lowest level of information. The sale units will be stored and stored in the outlet system at the unit level of product per transaction. The amount ordered is collected and stored by the customer at unit level per order in the order entry system. You can add up the individual transactions whenever you need the summary data. When you search for items in a product ordered this month and add them together, you have the total of all product orders entered in that month. Referral data is generally not tracked in an operating system. When a user requests an analysis in the data warehouse, the user has to view summary data first. A user can successfully promote the entire product unit throughout a given area. The user may then wish to examine the region's breakdowns. The next step might be to look at the following levels of the sales units in each store. The analysis often starts at a high level, and is reduced in detail.

Therefore, in a data store, the summary of data at different levels can be maintained effectively. You can provide answers at the simplest level or the most detailed level. The detail level of data is the level of granularity in an existing data warehouse. The more detail in the data, the finer the size of the data. You need to save a lot of permanent data in the data warehouse in order to safely store data. The type of granularity depends on how data will be processed, and the performance expectations. For example, each year details of each month, day, hour, minute, second, and so forth are available.

1.8 METADATA AND DATA WAREHOUSING

In a data warehouse, data is stored using a common schema controlled by a common dictionary. Within the data dictionary, data is kept about the logical data structures, file and address data, index information and others. The metadata should contain the following data warehouse information:

- The data structure based on the programmer's view
- Data structure based on DSS analysts' view
- The DW's data sources
- The data transformation at the moment of its migration to DW
- Model of data
- The connection between the data model and the DW
- Data extraction history

In the DW environment, metadata is a major component. Metadata helps to control reporting accuracy, validates the transformation of data and ensures calculation accuracy. Metadata also complies with the company end-users' definition of business terms. More details on metadata are provided in the next Unit.

1.9 DATA WAREHOUSING APPLICATIONS

In different sectors, there are numerous applications such as e-commerce, telecommunication, transport, marketing, distribution and retail. Given below are some of the applications of data warehouses:

Investment and Insurance: In this sector, data warehousing is used to analyze the customer, market trends and other patterns of data. The two sub-sectors where data warehousing plays an important role are Forex and stock markets.

Healthcare: A data warehousing system is used to forecast outcomes of a treatment generate its reports and share the data with different units. These units can be the research labs, medical units, and insurance providers. Enterprise data warehouses serve as the backbone of healthcare systems as they are updated with recent information which is crucial for saving lives.

Retail: Be it distribution, marketing, examining pricing policies, keeping a track of promotional deals, and finding the pattern in the customer buying trends: data warehousing solves it all. Many retail chains incorporate enterprise data warehousing for business intelligence and forecasting.

Social Media Websites: Social networking sites such as Facebook, Twitter, LinkedIn etc. are based on large data sets analyses. These sites collect data on members, groups; locations etc. and store this information in a single central repository. Data warehouse is necessary to implement the same data, because of its high volume of data.

Banking: Most banks are now using warehouses to see account/cardholder spending patterns. They use this to make special offers, deals, etc. available.

Government: In addition to store and analyze taxes used to detect tax theft, government uses the data warehouse.

Airlines: It is used in the airline system for operational purposes such as crew assignments, road profitability analyses, flight frequency programs promotions, etc.

Public sector: Information is collected in the public sector's data warehouse. It helps government agencies and departments manage their data and records.

1.10 TYPES OF DATA WAREHOUSES

There are three different types of traditional Data Warehouse models as listed below:

i. Enterprise

ii. Operational

iii. Data Mart

(i) Enterprise Data Warehouse

An enterprise provides a central repository database for decision support throughout the enterprise. It is a central place where all business information from different sources and applications are made available. Once it is stored, it can be used for analysis and used by all the people across the organization. The enterprise goal is to provide a complete overview of any particular object in the data model.

(ii) Operational Data Warehouse

These features have a sizable enterprise-wide scope, but unlike the substantial enterprise warehouse, data is refreshed in near real-time and used for routine commercial activity. It assists in obtaining data straight from the database, which also helps data transaction processing. The data present in the Operational Data Store can be scrubbed, and the duplication which is present can be reviewed and fixed by examining the corresponding market rules.

(iii) Data Mart

Data Mart may be a subset of knowledge warehouse, and it supports a specific region, business unit, or business function. Data Mart focuses on storing data for a particular functional area, and it contains a subset of data saved in a memory. Data Marts help in enhancing user responses and also reduces the volume of data for data analysis. It makes it more comfortable to go forward with the report. More on Data Marts can be studied in the next Unit.

1.11 POPULAR DATA WAREHOUSE PLATFORMS

A data warehouse is a critical database for supporting data analysis and acts as a conduit between analytical tools and operational data stores. The most popular data warehousing solutions include a range of useful features for data management and consolidation.

You can use them to extract/curate data from a range of environments, transform data and remove duplicates, and ensure consistency in your analytics.

Google BigQuery

BigQuery is a cost-effective data warehousing tool with built-in machine learning capabilities. You can integrate it with Cloud ML and TensorFlow to create powerful AI models. It can also execute queries on petabytes of data for real-time analytics. This scalable and serverless cloud data warehouse is ideal for companies that want to keep costs low. If you need a quick way to make informed decisions through data analysis, BigQuery is one of the solutions.

AWS Redshift

Redshift is a cloud-based data warehousing tool for enterprises. The platform can process petabytes of data quite fast. That's why it's suitable for high-speed data analytics. It also supports automatic concurrency scaling. The automation increases or decreases query processing resources to match workload demand.

Although tooling provided by Amazon reduces the need to have a database administrator full time, it does not eliminate the need for one. Amazon Redshift is known to have issues with handling storage efficiently in an environment prone to frequent deletes.

Snowflake

Snowflake is a data warehousing solution that offers a variety of options for public cloud technology. With Snowflake, you can make your business more data-driven. You may use Snowflake to set up an enterprise-grade cloud data warehouse. With Snowflake, you can analyze data from various unstructured and structured sources. However, Snowflake is dependent on Azure, Amazon Web Services (AWS), Google Cloud Services (GCS). The support can be a problem whenever one of those cloud servers has an independent outage.

Microsoft Azure Synapse

Microsoft Azure is a robust platform for data management, analytics, integration, and more, with solutions spanning AI, blockchain, and more than a dozen unique databases for varying use cases. Among them is Azure Synapse, formerly known as Azure SQL Data Warehouse, a platform built for analytics, providing you the ability to query data using either serverless or provisioned resources at scale.

Azure Synapse brings together the two worlds of data warehousing and analytics with a unified experience to ingest, prepare, manage, and serve data for immediate BI and machine learning. The broader Azure platform includes thousands of tools, including others that interface with the various Azure databases.

1.12 SUMMARY

In this unit you have studied about the evolution, characteristics, benefits and applications of Data Warehouse.

Operational database system provides day-to-day information, although strategic decision-making cannot be used easily. Data Warehouse is a concept designed to aid strategic information. Data Warehouse allows people to make decisions and provides flexible, convenient and interactive sources of strategic intelligence. A data warehouse combines several technologies because it collects data from various operational data base systems and external sources such as magazines, newspapers and reports from the same industry, removes contradictions, transforms the data and then stores them in formats suited to easy access for decision-making purposes. The defining characteristics of the data warehouse are: Subject oriented, integrated, time-variant, and non-volatile.

Data warehouses are meant to be used by executives, managers, and other people at higher managerial levels who may not have much technical expertise in handling the databases.

Advantages of data warehouses include better decisions, increased productivity, lower operational costs, enhanced asset and liability management, and better CRM.

1.13 SOLUTIONS/ANSWERS

☞ **Check Your Progress 1**

1) Data Warehousing (DW) is a process for collecting and managing data from diverse sources to provide meaningful insights into the business. A Data Warehouse is typically used to connect and analyze heterogeneous sources of business data. The data warehouse is the centerpiece of the BI system built for data analysis and reporting.

It is amalgam of technologies and components which helps to use data strategically. Instead of transaction processing, it is the automated collection of a vast amount of information by a company that is configured for demand and review. It's a process of transforming data into information and making it available for users to make a difference in a timely way.

The archive of decision support (Data Warehouse) is managed independently from the operating infrastructure of the organization. The data warehouse, however, is not a product but rather an environment. It is an organizational framework of an information system that provides consumers with knowledge regarding current and historical decision help that is difficult to access or present in the conventional operating data store.

Data storage platforms also sort data on a variety of subjects like customers, products or business.

- Data storage is a tool that companies can use increasingly important for corporate intelligence:
- Make uniformity possible. All research data gathered and shared to decision makers worldwide should be used in a uniform format. Standardization of data from various sources reduces the risk of misinterpretation as well as overall accuracy of interpretation.
- Take better business decisions. Successful entrepreneurs have a thorough understanding of data, and are good at predicting future trends. The data storage system helps users access various data sets at speed and efficiency.
- Data storage platforms allow companies to access their business' past history and evaluate ideas and projects. This gives managers an idea of how they can improve their sales and management practices.

2). Following are the four main characteristics of a data warehouse:

i) Subject oriented

A data warehouse is subject-oriented, as it provides information on a topic rather than the ongoing operations of organizations. Such issues may be inventory, promotion, storage, etc. Never does a data warehouse concentrate on the current processes. Instead, it emphasized modeling and analyzing decision-making data. It also provides a simple and succinct description of the particular subject by excluding details that would not be useful in helping the decision process.

(ii) Integrated

Integration in Data Warehouse means establishing a standard unit of measurement from the different databases for all the similar data. The data must also get stored in a simple and universally acceptable manner within the Data Warehouse. Through combining data from various sources such as a mainframe, relational databases, flat files, etc., a data warehouse is created. It must also keep the naming conventions, format, and coding consistent. Such an application assists in robust data analysis. Consistency must be maintained in naming conventions, measurements of characteristics, specification of encoding, etc.

(iii) Time-variant

Compared to operating systems, the time horizon for the data warehouse is given period and provides historical information. It contains a temporal element, either explicitly or implicitly.One such location in the record key system where Data Warehouse data shows time variation is. Each primary key contained with the DW should have an element of time either implicitly or explicitly. Just like the day, the month of the week, etc.

(iv) Non-volatile

Also, the data warehouse is non-volatile, meaning that prior data will not be erased when new data are entered into it. Data is read-only, only updated regularly. It also assists in analyzing historical data and in understanding what and when it happened. The transaction process, recovery, and competitiveness control mechanisms are not required. In the Data Warehouse environment, activities such as deleting, updating, and inserting that are performed in an operational application environment are omitted.

☞ Check Your Progress 2

1) Data Warehouse systems are segregated from production databases so that they aren't intermingled and cause conflicts.

   - There is a database available for tasks such as searching records, indexing, and digital archiving. Data warehouse queries are often complex due to their varied and complex nature.
   - It is possible to manage multiple transactions simultaneously through business databases. Concurrency control and recovery mechanisms are needed to ensure that the database in operational databases is robust and consistent.
   - The operational database query allows for reading and modification of operations, whilst the read access to stored information is required for OLAP queries only.
   - A database of operations maintains current information. In contrast, historical data is kept in a warehouse.

2) A database stores the current data required to power an application. A data warehouse stores current and historical data from one or more systems in a predefined and fixed schema, which allows business analysts and data scientists to easily analyze the data. The table below summarizes differences between databases, data warehouses:

Table 2: Database Vs Data Warehouse

Characteristic Feature	Database	Data Warehouse
Workloads	Operational and transactional	Analytical
Data Type	Structured or semi-structured	Structured and/or semi-structured
Schema Flexibility	Rigid or flexible schema depending on database type	Pre-defined and fixed schema definition for ingest (schema on write and read)
Data Freshness	Real time	May not be up-to-date based on frequency of ETL processes
Users	Application developers	Business analysts and data scientists
Pros	Fast queries for storing and updating data	The fixed schema makes working with the data easy for business analysts
Cons	May have limited analytics capabilities	Difficult to design and evolve schema
		Scaling compute may require unnecessary scaling of storage, because they are tightly coupled

1.14 FURTHER READINGS

1. William H. Inmon, Building the Data Warehouse, Wiley, 4th Edition, 2005.
2. Data Warehousing Fundamentals, Paulraj Ponnaiah, Wiley Student Edition.
3. Data Warehousing, Reema Thareja, Oxford University Press, 2011.

UNIT 2 DATA WAREHOUSE ARCHITECTURE

Structure

2.0 INTRODUCTION

In the previous unit we had studied about the data warehousing and related topics. Despite numerous advancements over the last five years in the arena of Big Data, cloud computing, predictive analysis, and information technologies, data warehouses have gained more significance. For the success of any data warehouse, its architecture plays an important role. Since three decades, the data warehouse architecture has been the pillar of the corporate data ecosystems.

This unit present various topics including the basic concept of data warehouse architecture, its types, significant components and layers of data ware house architecture, data marts and their designing.

2.1 OBJECTIVES

After going through this unit, you shall be able to:

- understand the purpose of data warehouse architecture;
- describe the process of storing the data in a data warehouse;
- list and discuss the various types of data warehouse architectures;
- discuss various components and layers of data warehouse architecture;
- to summarize the functionality of data marts, their benefits and various types, and
- to know the ways of structuring and designing the data marts.

2.2 DATA WAREHOUSE ARCHITECTURE AND ITS TYPES

Data warehouse architecture is a data storage framework's design of an organization. It takes information from raw data sets and stores it in a structured and easily digestible format.

A data warehouse architecture plays a vital role in the data enterprise. As databases assist in storing and processing data, and data warehouses help in analyzing that data.

Data warehousing is a process of storing a large amount of data by a business or organization. The data warehouse is designed to perform large complex analytical queries on large multi-dimensional datasets in a straightforward manner. Data warehouses extract data from different resources, which are in different fonts, convert it into a unique form, and place data in Data Warehouse.

2.2.1 Types of Data Warehouse Architectures

Data warehouse architecture defines the arrangement of the data in different databases. As the data must be organized and cleansed to be valuable, a modern data warehouse structure identifies the most effective technique of extracting information from raw data.

Using a dimensional model, the raw data in the staging area is extracted and converted into a simple consumable warehousing structure to deliver valuable business intelligence. When designing a data warehouse, there are three different types of models to consider, based on the approach of number of tiers the architecture has.

(i) Single-tier data warehouse architecture

(ii) Two-tier data warehouse architecture

(iii) Three-tier data warehouse architecture

The details of each of the architecture are given below:

(i) Single-tier data warehouse architecture

The single-tier architecture (Figure 1) is not a frequently practiced approach. The main goal of having such architecture is to remove redundancy by minimizing the amount of data stored. Its primary disadvantage is that it doesn't have a component that separates analytical and transactional processing.

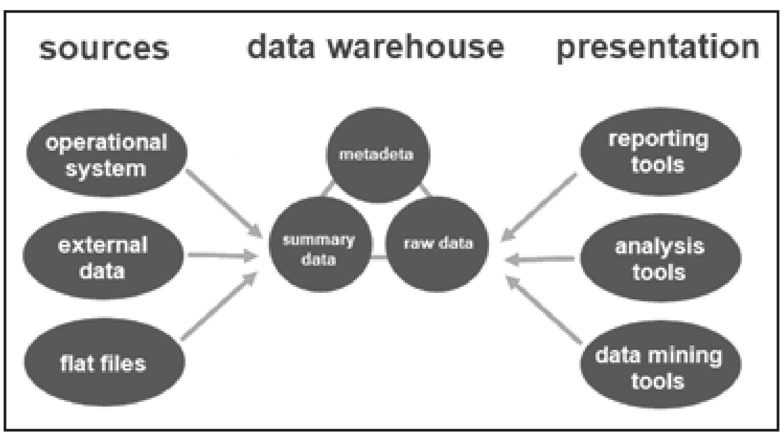

Figure 1: Single Tier Data Warehouse Architecture

(ii) Two-tier data warehouse architecture

The two-tier architecture (Figure 2) includes a staging area for all data sources, before the data warehouse layer. By adding a staging area between the sources and the storage repository, you ensure all data loaded into the warehouse is cleansed and in the appropriate format.

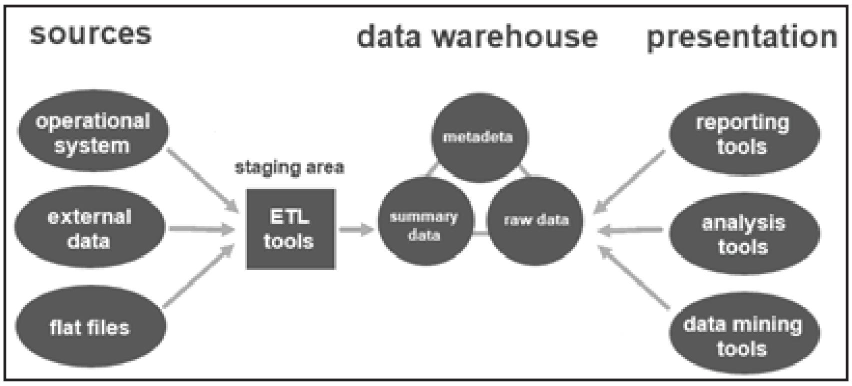

Figure 2: Two-Tier Data Warehouse Architecture

(iii) Three-tier data warehouse architecture

The three-tier approach (Figure 3) is the most widely used architecture for data warehouse systems.

Essentially, it consists of three tiers:

1. **The bottom tier** is the database of the warehouse, where the cleansed and transformed data is loaded.

2. **The middle tier** is the application layer giving an abstracted view of the database. It arranges the data to make it more suitable for analysis. This is done with an OLAP server, implemented using the ROLAP or MOLAP model.

3. **The top-tier** is where the user accesses and interacts with the data. It represents the front-end client layer. You can use reporting tools, query, analysis or data mining tools.

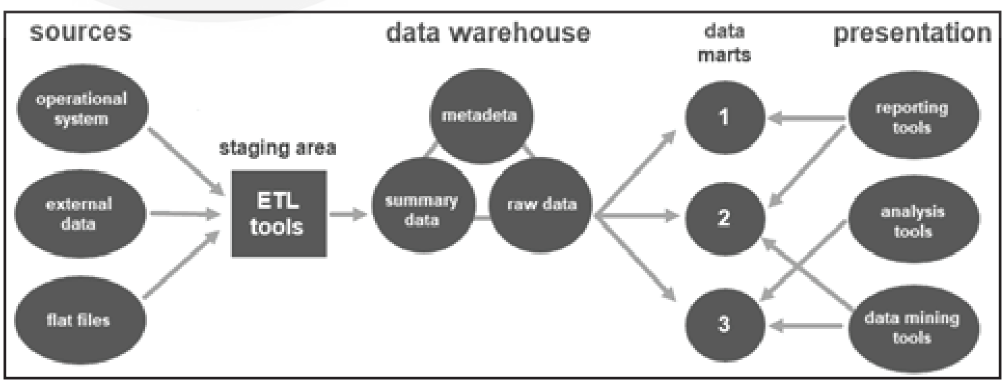

Figure 3: Three-Tier Data Warehouse Architecture

Figure 4 illustrates the complete data warehouse architecture with three tiers:

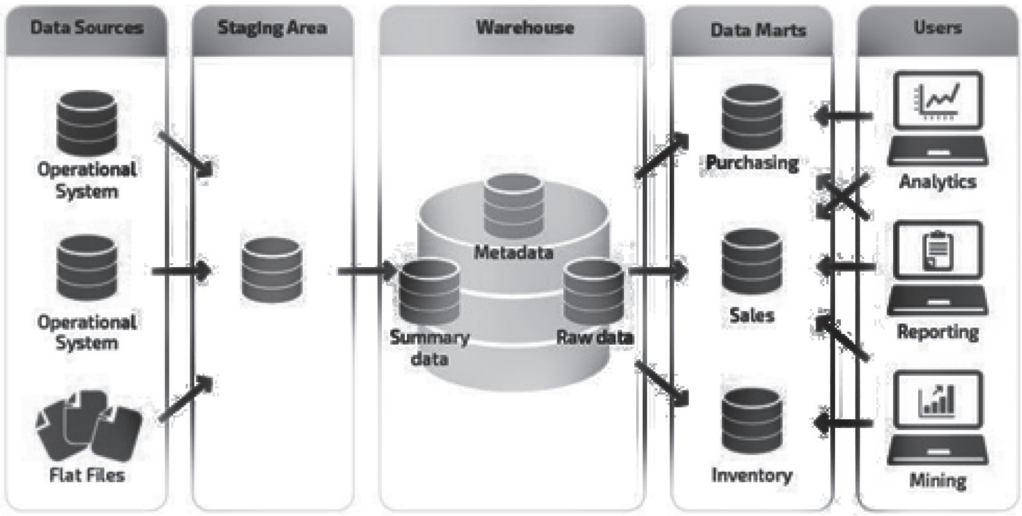

Figure 4: Three-Tier Data Warehouse Architecture

2.2.2 Cloud-based Data Warehouse Architecture

Cloud-based data warehouse architecture is relatively new when compared to legacy options. This data warehouse architecture means that the actual data warehouses are accessed through the cloud. There are several cloud based data warehouses options, each of which has different architectures for the same benefits of integrating, analyzing, and acting on data from different sources. The difference between a cloud-based data warehouse approach compared to that of a traditional approach include:

- **Up-front costs:** The different components required for traditional, on-premises data warehouses mandate pricey up-front expenses. Since the components of cloud architecture are accessed through the cloud, these expenses don't apply.
- **Ongoing costs:** While businesses with on-prem data warehouses must deal with upgrade and maintenance costs, the cloud offers a low, pay-as-you-go model.
- **Speed:** Cloud-based data warehouse architecture is substantially speedier than on-premises options, partly due to the use of ELT — which is an uncommon process for on-premises counterparts.
- **Flexibility:** Cloud data warehouses are designed to account for the variety of formats and structures found in big data. Traditional relational options are designed simply to integrate similarly structured data.
- **Scale:** The elastic resources of the cloud make it ideal for the scale required of big datasets. Additionally, cloud-based data warehousing options can also scale down as needed, which is difficult to do with other approaches.

Cloud-based platforms make it possible to create, share, and store massive data sets with ease, paving the way for more efficient and effective data access and analysis. Cloud systems are built for sustainable business growth, with many modern Software-as-a Service (SaaS) providers separating data storage from computing to improve scalability when querying data.

Some of the more notable cloud data warehouses in the market include Amazon Redshift, Google BigQuery, Snowflake, and Microsoft Azure SQL Data Warehouse.

Now, let's learn about the major components of a data warehouse and how they help build and scale a data warehouse in the next section.

2.3 COMPONENTS OF DATA WAREHOUSE ARCHITECTURE

A data warehouse design consists of six main components:

- Data Warehouse Database
- ETL
- Metadata
- Data Warehouse Access Tools
- Data Warehouse Bus
- Data Warehouse Reporting Layer

The details of all the components are given below:

2.3.1 Data Warehouse Database

The central component of DW architecture is a data warehouse database that stocks all enterprise data and makes it manageable for reporting. Obviously, this means you need to choose which kind of database you'll use to store data in your warehouse.

The following are the four database types that you can use:

- Typical relational databases are the row-centered databases you perhaps use on an everyday basis —for example, Microsoft SQL Server, SAP, Oracle, and IBM DB2.
- Analytics databases are precisely developed for data storage to sustain and manage analytics, such as Teradata and Greenplum.
- Data warehouse applications aren't exactly a kind of storage database, but several dealers now offer applications that offer software for data management as well as hardware for storing data. For example, SAP Hana, Oracle Exadata, and IBM Netezza.
- Cloud-based databases can be hosted and retrieved on the cloud so that you don't have to procure any hardware to set up your data warehouse—for example, Amazon Redshift, Google BigQuery, and Microsoft Azure SQL.

2.3.2 Extraction, Transformation, and Loading Tools (ETL)

ETL tools are central components of enterprise data warehouse architecture. These tools help extract data from different sources, transform it into a suitable arrangement, and load it into a data warehouse.

The ETL tool you choose will determine:

- The time expended in data extraction
- Approaches to extracting data

- Kind of transformations applied and the simplicity to do so
- Business rule definition for data validation and cleansing to improve end-product analytics
- Filling mislaid data
- Outlining information distribution from the fundamental depository to your BI applications

More on ETL can be studied in Unit-4

2.3.3 Metadata

Before we delve into the different types of metadata in data mining, we first need to understand what metadata is. In the data warehouse architecture, metadata describes the data warehouse database and offers a framework for data. It helps in constructing, preserving, handling, and making use of the data warehouse.

There are two types of metadata in data mining:

- Technical Metadata comprises information that can be used by developers and managers when executing warehouse development and administration tasks.
- Business Metadata comprises information that offers an easily understandable standpoint of the data stored in the warehouse.

Metadata plays an important role for businesses and the technical teams to understand the data present in the warehouse and convert it into information.

2.3.4 Data Warehouse Access Tools

A data warehouse uses a database or group of databases as a foundation. Data warehouse corporations generally cannot work with databases without the use of tools unless they have database administrators available. However, that is not the case with all business units. This is why they use the assistance of several no-code data warehousing tools, such as:

- Query and reporting tools help users produce corporate reports for analysis that can be in the form of spreadsheets, calculations, or interactive visuals.
- Application development tools help create tailored reports and present them in interpretations intended for reporting purposes.
- Data mining tools for data warehousing systematize the procedure of identifying arrays and links in huge quantities of data using cutting-edge statistical modeling methods.
- OLAP tools help construct a multi-dimensional data warehouse and allow the analysis of enterprise data from numerous viewpoints.

2.3.5 Data Warehouse Bus

It defines the data flow within a data warehousing bus architecture and includes a data mart. A data mart is an access level that allows users to transfer data. It is also used for partitioning data that is produced for a particular user group.

2.3.6 Data Warehouse Reporting Layer

The reporting layer in the data warehouse allows the end-users to access the BI interface or BI database architecture. The purpose of the reporting layer in the data warehouse is to act as a dashboard for data visualization, create reports, and take out any required information.

Constructing a data warehouse is primarily dependent on a particular business. And every data warehouse architecture has four layers. Let's study them in following section.

2.4 LAYERS OF DATA WAREHOUSE ARCHITECTURE

In general, the data warehouse architecture can be divided into four layers. They are:

i. Data Source Layer

ii. Data Staging Layer

iii. Data Storage Layer

iv. Data Presentation Layer

Let us study the various layers and their functionality.

(i) Data source layer

The data source layer is the place where unique information, gathered from an assortment of inner and outside sources, resides in the social database. Following are the examples of the data source layer:

- **Operational Data** — Product information, stock information, marketing information, or HR information
- **Social Media Data** — Website hits, content fame, contact page completion
- **Outsider Data** — Demographic information, study information, statistics information

While most data warehouses manage organized data, thought ought to be given to the future utilization of unstructured data sources, for example, voice accounts, scanned pictures, and unstructured text. These floods of data are significant storehouses of information and ought to be viewed when building up your warehouse.

(ii) Data Staging Layer

This layer dwells between information sources and the data warehouse. In this layer, information is separated from various inside and outer data sources. Since source data comes in various organizations, the data extraction layer will use numerous technologies and devices to extricate the necessary information. Once the extracted data has been stacked, it will be exposed to high-level quality checks. The conclusive outcome will be perfect and organized data that you will stack into your data warehouse. The staging layer contains the given parts:

- **Landing Database and Staging Area**

The landing database stores the information recovered from the data source. Before the data goes to the warehouse, the staging process does stringent quality checks on it. Arranging is a basic step in architecture. Poor information will add up to inadequate data, and the result is poor business dynamic. The arranging layer is where you need to make changes in accordance with the business process to deal with unstructured information sources.

- **Data Integration Tool**

Extract, Transform and Load tools (ETL) are the data tools used to extricate information from source frameworks, change, and prepare information and load it into the warehouse.

(iii) Data Storage Layer

This layer is the place where the data that was washed down in the arranging zone is put away as a solitary central archive. Contingent upon your business and your warehouse architecture necessities, your data storage might be a data warehouse center, data mart (data warehouse somewhat recreated for particular departments), or an Operational Data Store (ODS).

(iv) Data Presentation Layer

This is where the users communicate with the scrubbed and sorted out data. This layer of the data architecture gives users the capacity to query the data for item or service insights, break down the data to conduct theoretical business situations, and create computerized or specially appointed reports.

You may utilize an OLAP or reporting instrument with an easy to understand Graphical User Interface (GUI) to assist users with building their queries, perform analysis, or plan their reports.

2.4.1 Best Practices for Data Warehouse Architecture

Designing the data warehouse with the designated architecture is an art. Some of the best practices are shown below:

- Create data warehouse models that are optimized for information retrieval in both dimensional, de-normalized, or hybrid approaches.
- Select a single approach for data warehouse designs such as the top-down or the bottom-up approach and stick with it.
- Always cleanse and transform data using an ETL tool before loading the data to the data warehouse.
- Create an automated data cleansing process where all data is uniformly cleaned before loading.
- Allow sharing of metadata between different components of the data warehouse for a smooth retrieval process.
- Always make sure that data is properly integrated and not just consolidated when moving it from the data stores to the data warehouse. This would require the 3NF normalization of data models.
- Monitor the performance and security. The information in the data warehouse is valuable, though it must be readily accessible to provide value to the organization. Monitor system usage carefully to ensure that performance levels are high.
- Maintain the data quality standards, metadata, structure, and governance. New sources of valuable data are becoming available routinely, but they require consistent management as part of a data warehouse. Follow procedures for data cleaning, defining metadata, and meeting governance standards.

- Provide an agile architecture. As the corporate and business unit usage increases, they will discover a wide range of data mart and warehouse needs. A flexible platform will support them far better than a limited, restrictive product.
- Automate the processes such as maintenance. In addition to adding value to business intelligence, machine learning can automate data warehouse technical management functions to maintain speed and reduce operating costs.
- Use the cloud strategically. Business units and departments have different deployment needs. Use on-premise systems when required, and capitalize on cloud data warehouses for scalability, reduced cost, and phone and tablet access.

2.5 DATA MARTS

A data mart is a subset of a data warehouse focused on a particular line of business, department, or subject area. Data marts make specific data available to a defined group of users, which allows those users to quickly access critical insights without wasting time searching through an entire data warehouse. For example, many companies may have a data mart that aligns with a specific department in the business, such as finance, sales, or marketing.

2.5.1 Data Mart Vs Data Warehouse

Data marts and data warehouses are both highly structured repositories where data is stored and managed until it is needed. However, they differ in the scope of data stored: data warehouses are built to serve as the central store of data for the entire business, whereas a data mart fulfills the request of a specific division or business function. Because a data warehouse contains data for the entire company, it is best practice to have strictly control who can access it. Additionally, querying the data you need in a data warehouse is an incredibly difficult task for the business. Thus, the primary purpose of a data mart is to isolate—or partition—a smaller set of data from a whole to provide easier data access for the end consumers.

A data mart can be created from an existing data warehouse—the top-down approach—or from other sources, such as internal operational systems or external data. Similar to a data warehouse, it is a relational database that stores transactional data (time value, numerical order, reference to one or more object) in columns and rows making it easy to organize and access.

On the other hand, separate business units may create their own data marts based on their own data requirements. If business needs dictate, multiple data marts can be merged together to create a single, data warehouse. This is the bottom-up development approach.

In a nut-shell, following are the differences:

- Data mart is for a specific company department and normally a subset of an enterprise-wide data warehouse.
- Data marts improve query speed with a smaller, more specialized set of data.
- Data warehouses help make enterprise-wide strategic decisions, data marts are for department level, tactical decisions.

- Data warehouse includes many data sets and takes time to update, data marts handle smaller, faster-changing data sets.
- Data warehouse implementation can take many years, data marts are much smaller in scope and can be implemented in months.

2.6 BENEFITS OF DATA MARTS

Data marts are designed to meet the needs of specific groups by having a comparatively narrow subject of data. And while a data mart can still contain millions of records, its objective is to provide business users with the most relevant data in the shortest amount of time.

With its smaller, focused design, a data mart has several benefits to the end user, including the following:

- **Cost-efficiency:** There are many factors to consider when setting up a data mart, such as the scope, integrations, and the process to extract, transform, and load (ETL). However, a data mart typically only incurs a fraction of the cost of a data warehouse.
- **Simplified data access:** Data marts only hold a small subset of data, so users can quickly retrieve the data they need with less work than they could when working with a broader data set from a data warehouse.
- **Quicker access to insights:** Intuition gained from a data warehouse supports strategic decision-making at the enterprise level, which impacts the entire business. A data mart fuels business intelligence and analytics that guide decisions at the department level. Teams can leverage focused data insights with their specific goals in mind. As teams identify and extract valuable data in a shorter space of time, the enterprise benefits from accelerated business processes and higher productivity.
- **Simpler data maintenance:** A data warehouse holds a wealth of business information, with scope for multiple lines of business. Data marts focus on a single line, housing fewer than 100GB, which leads to less clutter and easier maintenance.
- **Easier and faster implementation:** A data warehouse involves significant implementation time, especially in a large enterprise, as it collects data from a host of internal and external sources. On the other hand, you only need a small subset of data when setting up a data mart, so implementation tends to be more efficient and include less set-up time.

2.7 TYPES OF DATA MARTS

There are three types of data marts that differ based on their relationship to the data warehouse and the respective data sources of each system.

- **Dependent data marts** are partitioned segments within an enterprise data warehouse. This top-down approach begins with the storage of all business data in one central location. The newly created data marts extract a defined subset of the primary data whenever required for analysis.
- **Independent data marts** act as a standalone system that doesn't rely on a data warehouse. Analysts can extract data on a particular subject or business

process from internal or external data sources, process it, and then store it in a data mart repository until the team needs it.

- **Hybrid data marts** combine data from existing data warehouses and other operational sources. This unified approach leverages the speed and user-friendly interface of a top-down approach and also offers the enterprise-level integration of the independent method.

2.8 STRUCTURE OF A DATA MART

A data mart is a subject-oriented relational database that stores transactional data in rows and columns, which makes it easy to access, organize, and understand. As it contains historical data, this structure makes it easier for an analyst to determine data trends. Typical data fields include numerical order, time value, and references to one or more objects.

Companies organize data marts in a multidimensional schema as a blueprint to address the needs of the people using the databases for analytical tasks. The three main types of schema:

Star

Star schema is a logical formation of tables in a multidimensional database that resembles a star shape. In this blueprint, one fact table—a metric set that relates to a specific business event or process—resides at the center of the star, surrounded by several associated dimension tables.

There is no dependency between dimension tables, so a star schema requires fewer joins when writing queries. This structure makes querying easier, so star schemas are highly efficient for analysts who want to access and navigate large data sets.

Snowflake

A snowflake schema is a logical extension of a star schema, building out the blueprint with additional dimension tables. The dimension tables are normalized to protect data integrity and minimize data redundancy.

While this method requires less space to store dimension tables, it is a complex structure that can be difficult to maintain. The main benefit of using snowflake schema is the low demand for disk space, but the caveat is a negative impact on performance due to the additional tables.

Data Vault

Data vault is a modern database modeling technique that enables IT professionals to design agile enterprise data warehouses. This approach enforces a layered structure and has been developed specifically to combat issues with agility, flexibility, and scalability that arise when using the other schema models. Data vault eliminates star schema's need for cleansing and streamlines the addition of new data sources without any disruption to existing schema.

2.9 DESIGNING THE DATA MARTS

Data marts guide important business decisions at a departmental level. For example, a marketing team may use data marts to analyze consumer behaviors, while sales staff could use data marts to compile quarterly sales reports. As these tasks happen within their respective departments, the teams don't need access to all enterprise data.

Typically, a data mart is created and managed by the specific business department that intends to use it. The process for designing a data mart usually comprises the following steps:

(i) Essential Requirements Gathering

The first step is to create a robust design. Some critical processes involved in this phase include collecting the corporate and technical requirements, identifying data sources, choosing a suitable data subset, and designing the logical layout (database schema) and physical structure.

(ii) Build/Construct

The next step is to construct it. This includes creating the physical database and the logical structures. In this phase, you'll build the tables, fields, indexes, and access controls.

(iii) Populate/Data Transfer

The next step is to populate the mart, which means transferring data into it. In this phase, you can also set the frequency of data transfer, such as daily or weekly. This usually involves extracting source information, cleaning and transforming the data, and loading it into the departmental repository.

(iv) Data Access

In this step, the data loaded into the data mart is used in querying, generating reports, graphs, and publishing. The main task involved in this phase is setting up a meta-layer and translating database structures and item names into corporate expressions so that non-technical operators can easily use the data mart. If necessary, you can also set up API and interfaces to simplify data access.

(v) Manage

The last step involves management and observation, which includes:

- Controlling ongoing user access.
- Optimization and refinement of the target system for improved performance.
- Addition and management of new data into the repository.
- Configuring recovery settings and ensuring system availability in the event of failure.

2.10 LIMITATIONS WITH DATA MARTS

Prospective builders of data warehouses are frequently advised to "start small" with a data mart and use that kernel to expand gradually into a full blown data warehouse. This approach to warehousing generally leads to failed projects for several reasons.

Sometimes the new data mart is so successful that the configuration is overrun by user demands. The databases grow too large too fast, response times become unacceptably long, and user frustration leads to searching for other ways to get the answers.

The more common reason for failure is that the data mart is immediately unsuccessful because it is designed in such a way that users are unable to retrieve the sort of information they want and need to extract from the data. Databases are highly denormalized to respond to a small set of canned queries; summaries, rather than detail data, comprise the database so that fine-grained exploratory data analysis is not possible; and support for ad hoc queries is either absent or so poor as to discourage users from bothering with them.

The very factors that frequently defeat data mart projects are also the most commonly recommended approaches to designing data marts and data warehouses in the popular data warehousing literature:

- Denormalization (dimensional modeling)
- Storing aggregates at the expense of detail data
- Skewing performance toward a small, preselected set of queries at the expense of all other exploratory analyses

☞ **Check your Progress 1**

1. Define data warehouse architecture.

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2. What is the correct flow of the data warehouse architecture?

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3. Mention some Data Mart Use Cases.

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2.11 SUMMARY

Data warehouse architecture is the design and building blocks of the modern data warehouse. In this unit we have studied the basic building blocks of the data warehouse, data warehouse architecture, its types, architecture models, data marts, designing of data marts and limitations.

In this next unit we will study about Dimensional Modeling.

2.12 SOLUTIONS / ANSWERS

☞ **Check Your Progress 1:**

1. The method for defining the entire architecture of data communication processing as well as the presentation that exists for end-clients is the data warehouse architecture. Every data warehouse is different, and each of them is characterized based on the standard vital components.

   In simple words, a data warehouse is an information system that consists of commutative and historical data from single or multiple sources. The process of reporting and analysis of data in the organizations is simplified with the help of different data warehousing concepts. There are different approaches to constructing a data warehouse architecture. Any approach is used based on the requirements of the organizations.

2. On every operational database, there are a certain fixed number of operations that have to be applied. There are different well-defined techniques for delivering suitable solutions. Data warehousing is found to be more effective when the correct flow of the data warehouse architecture is completely followed.

   The four different processes that contribute to a data warehouse are extracting and loading the data, cleaning and transforming the data, backing up and archiving the data, and carrying out the query management process by directing them to the appropriate data sources.

3. Data marts are used to solve specific organizational problems, especially those that are unique to one department. Typical use cases for a data mart include:

   Focused Analytics

   Analytics is perhaps the most common application of data marts. The data in these repositories is entirely relevant to the requirements of the business department, with no extraneous information, resulting in faster and more accurate analysis. For example, financial analysts will find it easier to work with a financial data mart, rather than working with an entire data warehouse.

   Fast Turnaround

   Data marts are generally faster to develop than a data warehouse, as the developers are working with fewer sources and a limited schema. Data marts are ideal for data projects operating under challenging time constraints.

Permission Management

Data marts can be a risk-free way to grant limited data access without exposing the entire data warehouse. For example, dependent data mart contains a segment of warehouse data, and users are only able to view the contents of the mart. This prevents unauthorized access and accidental writes.

Better Resource Management

Data marts are sometimes used where there is a disparity in resource usage between different departments. For example, the logistics department might perform a high volume of daily database actions, which causes the marketing team's analytics tools to run slow. By providing each department with its own data mart, it's easier to allocate resources according to their needs.

2.13 FURTHER READINGS

1. William H. Inmon, Building the Data Warehouse, Wiley, 4th Edition, 2005.
2. Data Warehousing Fundamentals, Paulraj Ponnaiah, Wiley Student Edition, 2001.
3. Data Warehousing, Reema Thareja, Oxford University Press, 2011.

UNIT 3 DIMENSIONAL MODELING

Structure

3.0 INTRODUCTION

In the earlier unit, we had studied about the Data Warehouse Architecture and Data Marts. In this unit let us focus on the modeling aspects. In this unit we will go through the dimensional modeling, star schema, snowflake schema, aggregate tables and Fact constellation schema.

3.1 OBJECTIVES

After going through this unit, you shall be able to:

- understand the purpose of dimension modeling;
- identifying the measures, facts, and dimensions;
- discuss the fact and dimension tables and their pros and cons;
- discuss the Star and Snowflake schemas;
- explore comparative analysis of star and snowflake schema;
- describe Aggregate facts, fact constellation, and
- discuss various examples of star and snowflake schema.

3.2 DIMENSIONAL MODELING

Dimensional modeling is a data model design adopted when building a data warehouse. Simply, it can be understood that dimension modeling reduces the response time of query fired unlike relational systems. The concept behind dimensional modeling is all about the conceptual design. Firstly let's see the introduction to dimensional modeling and how it is different from a traditional data model design. A data model is a representation of how data is stored in a database and it is usually a diagram of the few tables and the relationships that exist between them. This modeling is designed to read, summarize and compute some numeric data from a data warehouse. A data warehouse is an example of a system that requires small number of large tables. This is due to many users using the application to read lot of data a characteristic of a data warehouse is to write the data once and read it many times over so it is the read operation that is dominant in a data warehouse. Now let's look at the data warehouse containing customer related information in a single table this makes it a lot easier for analytics just to count the number of customers by country but this time the use of tables in the data warehouse simplify the query processing. The main objective of dimension modeling is to provide an easy architecture for the end user to write queries and also, to reduce the number of relationships between the tables and dimensions hence providing efficient query handling.

Dimensional modeling populates data in a cube as a logical representation with OLAP data management. The concept was developed by Ralph Kimball. It has "fact" and "dimension" as its two important measure. The transaction record is divided into either "facts", which consists of business numerical transaction data, or "dimensions", which are the reference information that gives context to the facts. The more detail about fact and dimension is explained in the subsequent sections.

The main objective of dimension modeling is to provide an easy architecture for the end user to write queries. Also it will reduce the number of relationships between the tables and dimensions, hence providing efficient query handling.

The following are the steps in Dimension Modeling as shown in figure1.

1. Identify Business Process
2. Identify Grain (level of detail)
3. Identify dimensions and attributes
5. Build Schema

The model should describe the Why, How much, When/Where/Who and What of your business process.

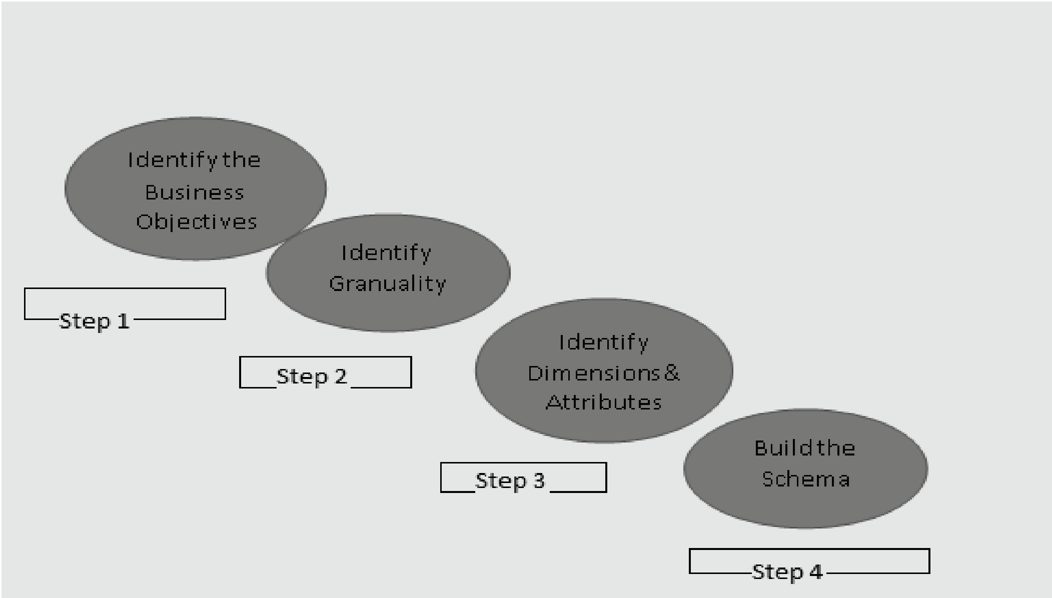

Figure 1: Steps in Dimension Modeling

Step 1: Identify the Business Objectives

Selection of the right business process to build a data warehouse and identifying the business objectives is the first step in dimension modeling. This is very important step otherwise this can lead to repeated process and software defects.

Step 2: Identifying Granularity

The grain literally means each minute detail of the business problem. This is decomposing of the large and complex problem into the lowest level information. For example, if there is some data month-wise. So, the table would contain details of all the months in a year. It depends on the report to be submitted to the management. This affects the size of the data warehouse.

Step 3: Identifying Dimensions and attributes

The dimensions of the data warehouse can be understood by the entities of the database. like, items, products, date, stocks, time etc. The identification of the primary keys and the foreign keys specifications all are described here.

Step 4: Build the Schema

The database structure or arrangement of columns in a database table, decides the schema. There are various popular schemas like, star, snowflake, fact constellation schemas - summarizing, from the selection of business process to identifying each and every finest level of detail of the business transactions. Identifying the significant dimensions and attributes would help to build the schema.

3.2.1 Strengths of Dimensional Modeling

Following are some of the strengths of Dimensional Modeling:

- It provides the simplicity of architecture or schema to understand and handle various stakeholders from warehouse designers to business clients.
- It reduces the number of relationships between different data elements.
- It promotes data quality by enforcing foreign key constraints as a form of referential integrity check on a data warehouse. The dimensional modeling

helps the database administrators to maintain the reliability of the data.

- The aggregate functions used in the schemas optimize the query performance posted by the customers. Since data warehouse size keeps on increasing and with this increased size, the optimization becomes the concern which dimension modeling makes it easy.

3.3 IDENTIFYING FACTS AND DIMENSIONS

We have studied the steps of dimension modeling in the previous section. The last step narrated is to build the schema. So, let's see the elementary measures to build a schema.

Facts and Fact table: A fact is an event. It is a measure which represents business items or transactions of items having association and context data. The Fact table contains the description of all the primary keys of all the tables used in the business processes which acts as a foreign key in the fact table. It also has an aggregate function to compute the business process on some entity. It is a numeric attribute of a fact, representing the performance or behavior of the business relative to the dimensions. The number of columns in the fact table is less than the dimension table. It is more normalized form.

Dimensions and Dimension table: It is a collection of data which describe one business dimension. Dimensions decide the contextual background for the facts, and they are the framework over which OLAP is performed. Dimension tables establish the context of the facts. The table stores fields that describe the facts. The data in the table are in de normalized form. So, it contains large number of columns as compared to fact table. The attributes in a dimension table are used as row and column headings in a document or query results display.

Example: In the example of student registration case study to any particular course can have attributes like student_id, course_id, program_id, date_of_registration, fee_id in fact table. Course summary can have course name, duration of the course etc. Student information can contain the personal details about the student like name, address, contact details etc.

Student Registration

Fact Table (student_id, course_id, program_id, date_of_registration, fee_id)
Measure: Sum (Fee_amount))
Dimension Tables (Student_details,
Course_details
Program_details,
Fee_details,
Date)

3.4 STAR SCHEMA

There are three basic popular models which are used for dimensional modeling:

- **Star Model**
- **Snowflake Model**
- **Fact Constellation Schema**

Star Schema: It represents the multidimensional model. In this model the data is organized into facts and dimensions. The star model is the underlying structure for a dimensional model. It has one broad central table (fact table) and a set of smaller tables (dimensions) arranged in a star design. This design is logically shown in the below figure 2.

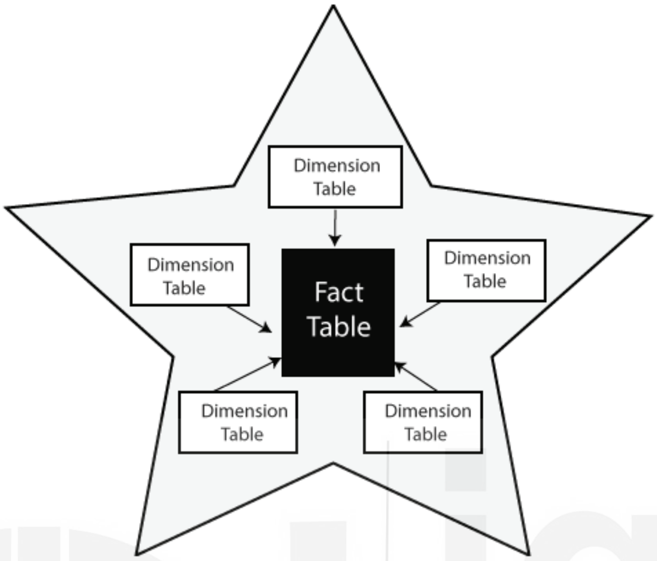

Figure 2 : Star Schema

3.4.1 Features of Star Schema

- The data is in denormalized database.
- It provides quick query response
- Star schema is flexible can be changed or added easily.
- It reduces the complexity of metadata for developers and end users.

3.5 ADVANTAGES AND DISADVANTAGES OF STAR SCHEMA

3.5.1 Advantages of Star Schema

Star schemas are easy for end users and applications to understand and navigate. With a well-designed schema, users can quickly analyze large, multidimensional data sets. The main advantages of star schemas in a decision-support environment are:

- **Query performance**

Because a star schema database has a small number of tables and clear join paths, queries run faster than they do against an OLTP system. Small single-table queries, usually of dimension tables, are almost instantaneous. Large join queries that involve multiple tables take only seconds or minutes to run.

In a star schema database design, the dimensions are linked only through the central fact table. When two dimension tables are used in a query, only one join path, intersecting the fact table, exists between those two tables. This design feature enforces accurate and consistent query results.

- **Load performance and administration**

Structural simplicity also reduces the time required to load large batches of data into a star schema database. By defining facts and dimensions and separating them into different tables, the impact of a load operation is reduced. Dimension tables can be populated once and occasionally refreshed. You can add new facts regularly and selectively by appending records to a fact table.

- **Built-in referential integrity**

A star schema has referential integrity built in when data is loaded. Referential integrity is enforced because each record in a dimension table has a unique primary key, and all keys in the fact tables are legitimate foreign keys drawn from the dimension tables. A record in the fact table that is not related correctly to a dimension cannot be given the correct key value to be retrieved.

- **Easily understood**

A star schema is easy to understand and navigate, with dimensions joined only through the fact table. These joins are more significant to the end user, because they represent the fundamental relationship between parts of the underlying business. Users can also browse dimension table attributes before constructing a query.

3.5.2 Disadvantages of Star Schema

As mentioned before, improving read queries and analysis in a star schema could involve certain challenges:

- *Decreased data integrity*: Because of the denormalized data structure, star schemas do not enforce data integrity very well. Although star schemas use countermeasures to prevent anomalies from developing, a simple insert or update command can still cause data incongruities.
- *Less capable of handling diverse and complex queries*: Databases designers build and optimize star schemas for specific analytical needs. As denormalized data sets, they work best with a relatively narrow set of simple queries. Comparatively, a normalized schema permits a far wider variety of more complex analytical queries.
- *No Many-to-Many Relationships*: Because they offer a simple dimension schema, star schemas don't work well for "many-to-many data relationships"

Example 1: Suppose a star schema is composed of a Sales fact table as shown in Figure 3a and several dimension tables connected to it for Time, Branch, Item and Location.

Fact Table

Sales is the Fact table.

Dimension Tables

The Time table has a column for each day, month, quarter, year etc..

The **Item** table has columns for each item_key, item_name, brand, type and supplier_type.

The **Branch** table has columns for each branch_key, branch_name and branch_type.

The **Location** table has columns of geographic data, including street, city, state, and country. Unit_Sold and Dollars_Sold are the Measures.

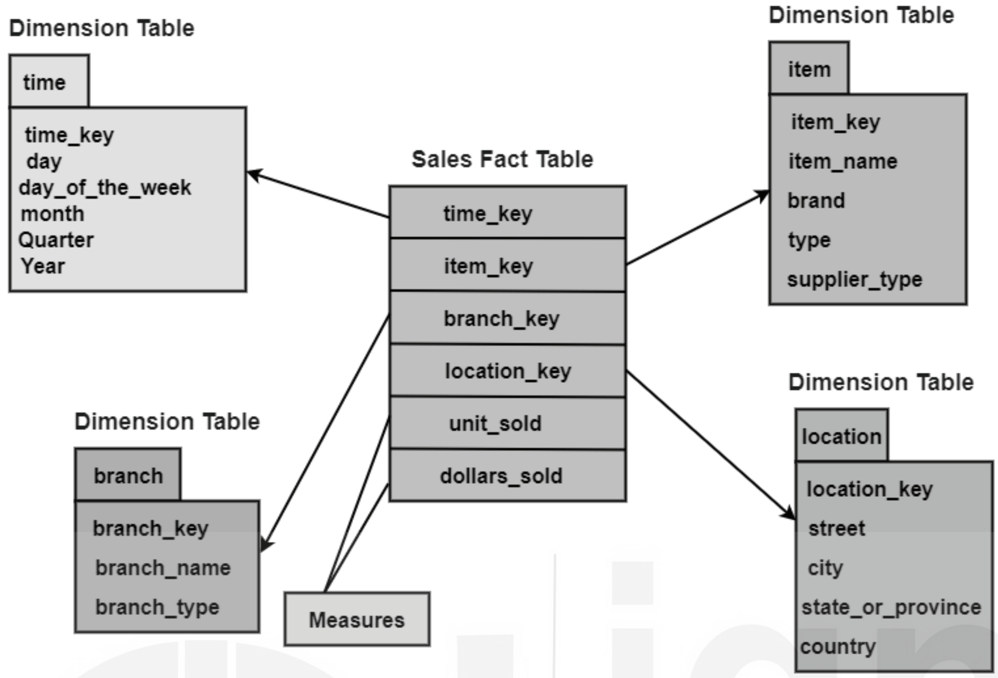

Figure 3a: Example of Star Schema

The measures may be unit_sold and dollars_sold.

Example 2:

The star schema works by dividing data into measurements and the "who, what, where, when, why, and how" descriptive context. Broadly, these two groups are facts and dimensions.

By doing this, the star schema methodology allows the business user to restructure their transactional database into smaller tables that are easier to fit together. Fact tables are then linked to their associated dimension tables with primary or foreign key relationships. An example of this would be a quick grocery store purchase. The amount you spent and how many items you bought would be considered a fact, but what you bought, when you bought it and the specific grocery store's location would all be considered dimensions.

Once these two groups have been established, we can connect them by the unique transaction number associated with your specific purchase. An important note is that each fact, or measurement, will be associated with multiple dimensions. This is what forms the star shape, the fact in the center, and dimensions drawing out around it. Dimensions relating to the grocery store, the products you bought, and descriptions about you as their customer will be carefully separated into its table with its attributes.

This example is modeled as shown below and star schema for this is depicted in Figure 3b.

Fact Table

Sales is the Fact Table.

Dimension Tables

The **Store** table consists of columns like store_id store_address, city, region, state and country.

Customer table has columns for each product_id, product_time and product_type.

Sales_Type includes sales_type_id and type_name columns.

Product table consists of product_id, product_name and product_type.

Time table consists of columns like time_id, action_date, action_week, action_month, action_year and action_ weekday.

Measures may be amount spent and no. of items bought.

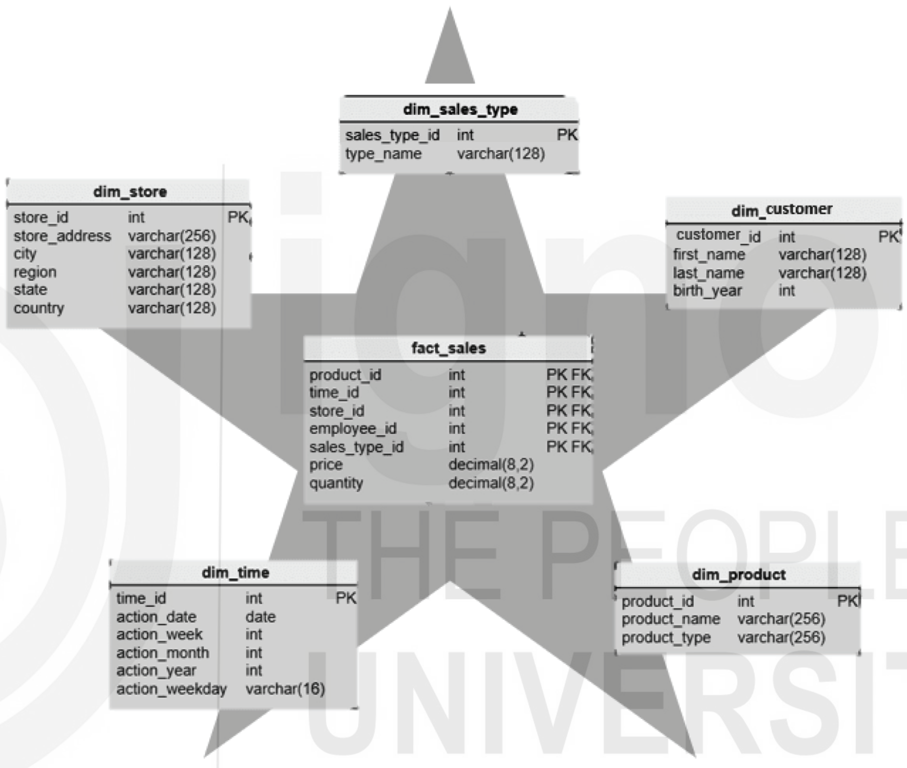

Figure 3b: Example of Star Schema

Check Your Progress 1

1) Discuss the characteristics of star schema?

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2) Draw a Star Schema for a marketing employee staying in a NewYork city of the country USA. He buys products and wants to compute the total product sold and how much sales done?

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3.6 SNOWFLAKE SCHEMA

The other popular modeling technique is Snowflake Schema. You can understand the term flakes as chocolate flakes on the pastry and ice-creams. These flakes add additional tastes to the chocolate. Similarly, snowflake schema is the extension of star schema which adds more dimensions to give more meaning to the logical view of the database. These additional tables are more normalized than star schema. The arrangement of data is like that the centralized fact table relates to multiple related dimensional tables. This can become more complex if the dimensions are more detailed and at multiple levels. In the conceptual hierarchy child table has multiple parent tables. You must keep in mind that we are just extending or flaking the dimension tables not the fact tables.

Snowflake Model

The snowflake model is the conclusion of decomposing one or more of the dimensions. Snowflake Schema in data warehouse is a logical arrangement of tables in a multidimensional database such that the ER diagram resembles a snowflake shape. A Snowflake Schema is an extension of a Star Schema, and it adds additional dimensions. The dimension tables are normalized which splits data into additional tables.

In the following Snowflake Schema example, Country is further normalized into an individual table.

3.6.1 Features of Snowflake Schema

Following are the important features of snowflake schema:

1. It has normalized tables
2. Occupy less disk space.
3. It requires more lookup time as many tables are interconnected and extending dimensions.

Example

In the figure 4, the snowflake schema is shown of a case study of customers, sales, products, location wise quantity sold, and number of items sold are calculated. The customers, products, date, store are saved in the fact table with their respective primary keys acting in fact table as a foreign key.

You will observe that the two aggregate functions can be applied to calculate quantity sold and amount sold. Further, the some dimensions are extended to the type of customer and also store information territory wise too. Note, date has been expanded into date, month, year. This schema will give you more opportunity to perform query handling in detail.

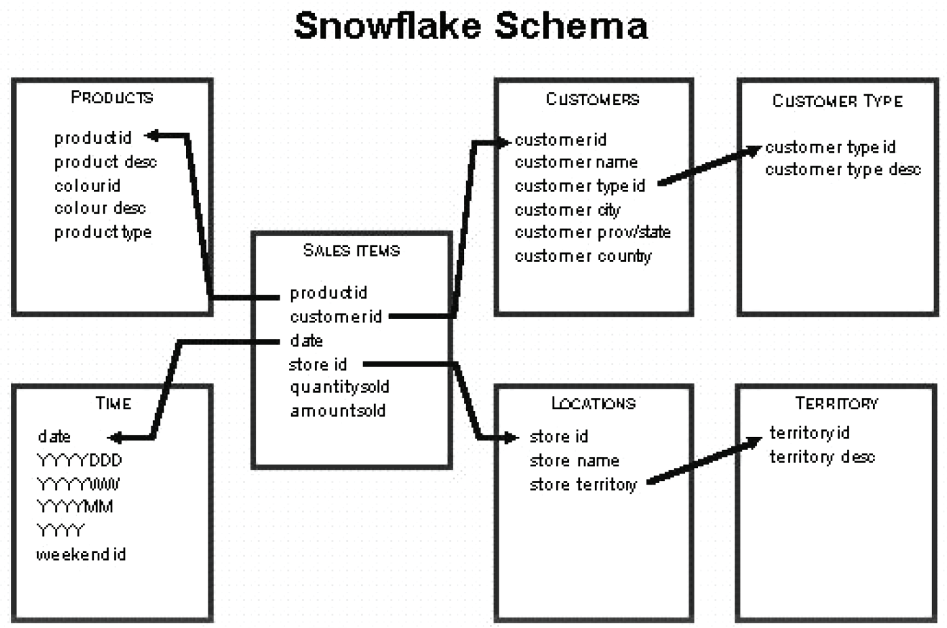

Figure 4: Snowflake Schema

3.7 ADVANTAGES AND DISADVANTAGES OF SNOWFLAKE SCHEMA

Following are the advantages of Snowflake schema:

- A Snowflake schema occupies a much smaller amount of disk space compared to the Star schema. Lesser disk space means more convenience and less hassle.
- Snowflake schema of small protection from various Data integrity issues. Most people tend to prefer the Snowflake schema because of how safe if it is.
- Data is easy to maintain and more structured.
- Data quality is better than star schema.

Disadvantages of Snowflake Schema

- Complex data schemas: As you might imagine, snowflake schemas create many levels of complexity while normalizing the attributes of a star schema. This complexity results in more complicated source query joins. In offering a more efficient way to store data, snowflake can result in performance declines while browsing these complex joins. Still, processing technology advancements have resulted in improved snowflake schema query performance in recent years, which is one of the reasons why snowflake schemas are rising in popularity.
- Slower at processing cube data: In a snowflake schema, the complex joins result in slower cube data processing. The star schema is generally better for cube data processing.
- Lower data integrity levels: While snowflake schemas offer greater normalization and fewer risks of data corruption after performing UPDATE and INSERT commands, they do not provide the level of transnational assurance that comes with a traditional, highly-normalized database

structure. Therefore, when loading data into a snowflake schema, it's vital to be careful and double-check the quality of information post-loading.

3.7.1 Star Schema Vs Snowflake Schema

Following are the differences between Star and Snowflake schema.

Features	Star Schema	Snowflake Schema
Normalized Dimension Tables	The dimension tables in star schema are not normalized so they may contain redundancies	This schema has normalized dimension tables
Queries	The execution of queries is relatively faster as there are less joins needed in forming a query.	The execution of snowflake schema complex queries is slower than star schema as many joins and foreign key relations are needed to form a query. Thus performance is affected.
Performance	Star schema model has faster execution and response time	It has slow performance as compared to star schema
Storage Space	This type of schema requires more storage space as compared to snowflake due to unnormalised tables.	Snowflake schema tables are easy to maintain and save storage space due to normalized tables.
Usage	Star schema is preferred when the dimension tables have lesser rows	If the dimension table contains large number of rows, snowflake schema is preferred
Type of DW	This schema is suitable for 1:1 or 1: many relationships such as data marts.	It is used for complex relationships such as many: many in enterprise Data warehouses.
Dimension Tables	Star schema has a single table for each dimension	Snowflake schema may have more than one dimension table for each dimension.

3.8 FACT CONSTELLATION SCHEMA

There is another schema for representing a multidimensional model. This term fact constellation is like the galaxy of universe containing several stars. It is a collection of fact schemas having one or more-dimension tables in common as shown in the Figure 5 below. This logical representation is mainly used in designing complex database systems.

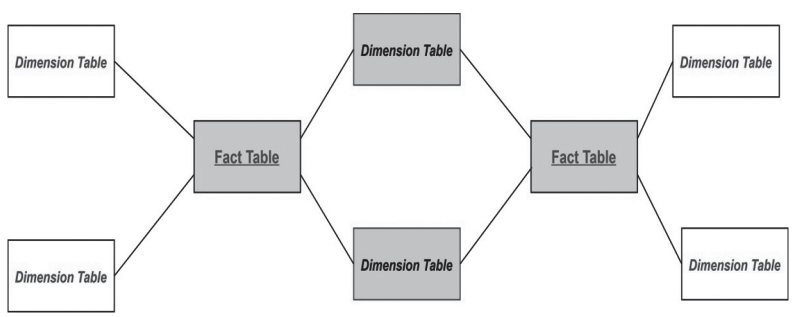

Figure 5: Fact Constellation Schema

In the figure 5 it can be observed that there are two fact tables and two-dimension tables in the pink boxes are the common dimension tables connecting both the star schemas.

For example, if we are designing a fact constellation schema for Placement and Workshop in a University consider,

Fact tables

Placement (Stud_roll, Company_id, TPO_id), need to calculate the number of students eligible and number of students placed.

Workshop (Stud_roll, Institute_id, TPO_id) need to find out the facts about number of students selected, number of students attended the workshop)

So, there are two fact tables namely, Placement and Workshop which are part of two different star schemas having:

i) dimension tables – Company, Student and TPO in Star schema with fact table Placement and

ii) dimension tables – Training Institute, Student and TPO in Star schema with fact table Workshop.

Both the star schema has two-dimension tables common and hence, forming a fact constellation or galaxy schema as shown in figure 6.

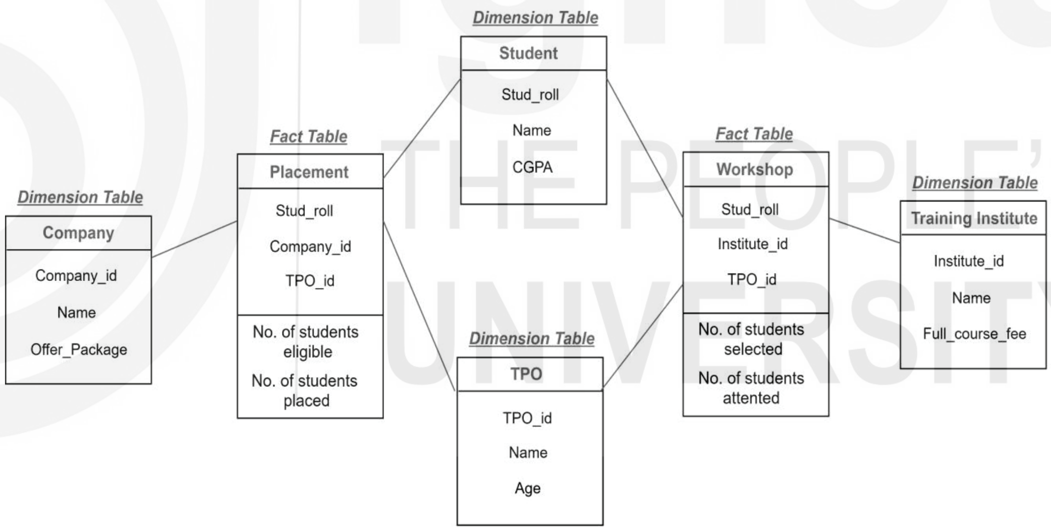

Figure 6: Fact Constellation of placement and workshop

3.8.1 Advantages and Disadvantages of Fact Constellation Schema

Advantage

This schema is more flexible and gives wider perspective about the data warehouse system.

Disadvantage

As, this schema is connecting two or more facts to form a constellation. This kind of structure makes it complex to implement and maintain.

☞ **Check Your Progress 2**

1. Compare and contrast Star schema with Snowflake Schema?

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2. Suppose that a data warehouse consists of dimensions time, doctor, ward and patient, and the two measures count and charge, where charge is the fee that a doctor charges a patient for a visit. Enumerate three classes of schemes that are popularly used for modeling.

   a) Draw a Star Schema diagram

   b) Draw a Snowflake Schema diagram.

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3.9 AGGREGATE TABLES

Since, in the data warehouse the data is stored in multidimensional cube. In the information technology industry, there are various tools available to process the queries posted on the data warehouse engine. These tools are called business intelligence (BI) tools. These tools help to answer the complex queries and to take decisions. Aggregate word is very similar to the aggregation of the database schemas of relational tables that you must be familiar with. Aggregate fact tables roll up the basic fact tables of the schema to improve the query processing. The business tools smoothly select the level of aggregation to improve the query performance. Aggregate fact tables contain foreign keys referring to dimension tables.

Points to note about Aggregate tables:

1) It is also called summary tables.

2) It contains pre-computed queries of the data warehouse schema.

3) It reduces the dimensionality of the base fact tables.

4) It can be used to respond to the queries of the dimensions that are saved.

3.10 NEED FOR BUILDING AGGREGATE FACT TABLES

Let us understand the need of building aggregate table. Aggregate tables also referred to pre-computed tables having partially summarized data.

- Simply putting in one word, it's about speed or quick response to queries. This you can understand as an intermediate table which stores the results of the queries on I/O disk space. It uses aggregates functionality.

  For example, there is a company ABC corporation limited which takes orders online and it there are millions of customer transactions placing orders. So, the dimension tables for the company could be Customer, Product and Order_date. In the fact table it maintains all the orders placed say, Fact_Orders. To generate a report of monthly orders by product type and by a particular region. It needs aggregates which are summary tables can be obtained by Groupby SQL query.

- It occupies less space than atomic fact tables. It nearly takes the half time of a general query processing.
- One of the more popular uses of aggregates is to adjust the granularity of a dimension. When the granularity of a dimension is changed, the fact table must be partially summarized to match the current grain of the new dimension, resulting in the creation of new dimensional and fact tables that fit this new grain standard.
- The Roll-up OLAP operation of the base fact tables generates aggregate tables. Hence the query performance increases as it reduces the number of rows to be accessed for the retrieval of data of a query.

3.11 AGGREGATE FACT TABLE AND DERIVED DIMENSION TABLES

Aggregate facts are produced by calculating measures from more atomic fact tables. These tables contain computational SQL aggregate functions like AVERAGE, MIN, MAX, COUNT etc. It also contains function that helps to find output using group by. The aggregate fact tables produce summary statistics. Whenever, the speedy query handling is required the aggregate fact tables is the best option.

- Basically, aggregates allow you to store the intermediate results or pre-calculate the subqueries or queries fired on a data warehouse by summing data up to higher levels and storing them in a separate star.
- You can understand aggregate fact tables as the conformed copy of the fact table as it should provide you the same result of the query as the detailed fact table.
- This aggregate fact tables can be used in the case of large datasets or when there are large number of queries. It reduces the response time of the queries fired by users or customers. It is very useful in business intelligence application tools.

When you have complicated questions of multiple facts in multiple tables that are stored at different levels from one another, and when a reporting request includes yet another level, the levels at which facts are stored become even more relevant. You must be able to meet users' need for fact reporting at the business level. There's nothing wrong with improving the overall intelligence.

The levels at which facts are stored become especially important when you begin to have complex queries with multiple facts in multiple tables that are stored at levels different from one another, and when a reporting request involves still a different level. You must be able to support fact reporting at the business levels which users require. There is nothing wrong with enhancing an aggregate with new facts or deriving new dimension. For measures, the only issue is if the new measures are atomic in the context of the aggregate fact. If, however, the new measures are received at a lower grain, you would be better off creating a new atomic fact for those measures prior to incorporating summarized measures into the aggregate. This would allow the new measures to be used for other purposes without having to go back to the source.

Let's say we have a fact table: FactBillReciept has monthly transactions. There can be different types of transaction receipts during a month for each supplier. This huge data would result in lot of calculations. So, we would build another aggregate table which is derived of base table.

FactBillMonthReceipt: It contains aggregated receipts per month, per supplier. But the problem is it has additional foreign keys like supplier_status for the month. To solve this, we have the concept of derived tables which contains additional measures and foreign keys that are not present in the base fact table.

Conformed Dimension

A conformed dimension is the dimension that is shared across multiple data mart or subject area. An organization may use the same dimension table across different projects without making any changes to the dimension tables.

Derived Tables

It is the significant addition to the Data Warehouse. Derived tables are used to create a second-level data marts for cross functional analysis.

Consolidated Fact tables: It is the fact table which has data from different fact tables used to form a schema with a common grain.

For example, to design a Sales department Data Warehouse schema assuming there are following entities and respective grains in them.

Sales: Employee, date, and product.

Budget: Department, Financial Year, Quarter-wise

Product can have various attributes like, product size, product _category etc..

One thing to notice here is that the product attributes keep on changing as per the requirements, but product dimension remains the same. So, it is better to keep Product as a separate dimension.

Let’s design the tables and its grains.

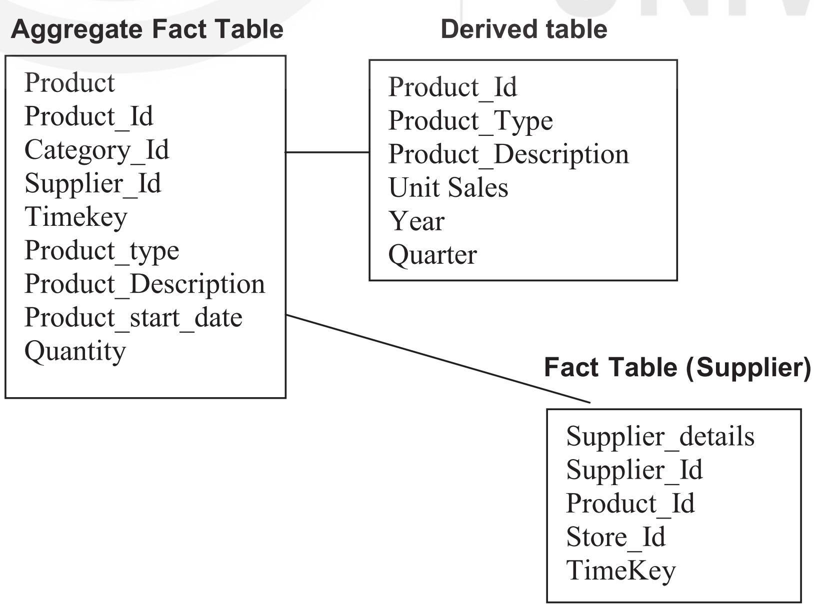

Figure 7: Aggregate Tables and Derived tables

The derived tables are very useful in terms of putting fewer loads on the Data Warehouse engine for calculation.

☞ **Check Your Progress 3**

1. Discuss the limitations of Aggregate Fact tables.

………………………………………………………………………………
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3.12 SUMMARY

This unit presented the basic designing of data warehouse. These topics are more focused on the various kind of modeling and schemas. It explored the grains, facts, and dimensions of the schemas. It is important to know about the dimensional modeling .as the appropriate modeling technique would yield the correct respond the queries.

A dimensional modeling is a kind of data structure used to optimize design of Data warehouse for the query retrieval operations. There are various schema designs. Here, it discussed star, snowflake, and fact constellations. From denormalized to normalized schemas uses dimension, fact, derived and aggregate fact table. Every table has some purpose and used for efficient designing in terms of space and query handling. This unit discusses the pros and cons of every tables. The number of examples used to explain the designing in different scenarios.

3.13 SOLUTIONS/ANSWERS

Check Your Progress 1:

1) Characteristics of Star Schema:

- Every dimension in a star schema is represented with only one-dimension table.
- The dimension table should contain the set of attributes.
- The dimension table is joined to the fact table using a foreign key
- The dimension table are not joined to each other
- Fact table would contain key and measure
- The Star schema is easy to understand and provides optimal disk usage.
- The dimension tables are not normalized. For instance, in the above figure, Country ID does not have Country lookup table as an OLTP design would have.
- The schema is widely supported by BI Tools

2)

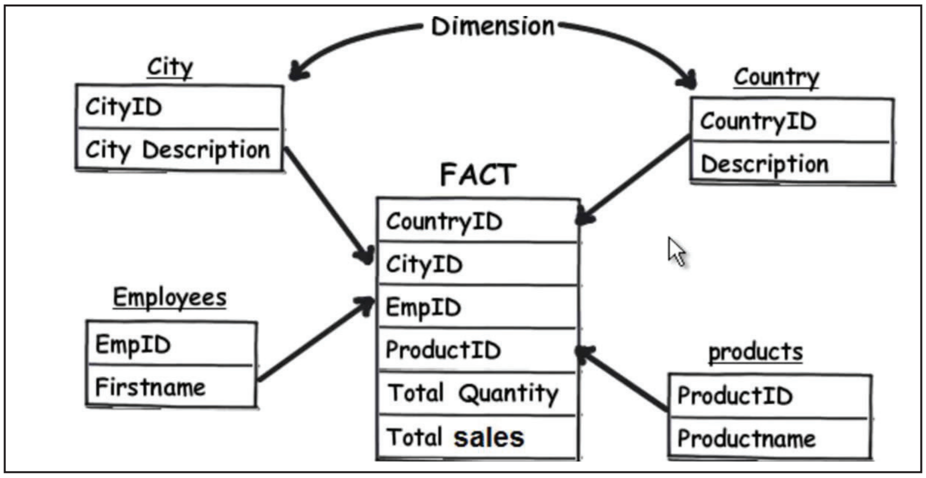

Figure 8: Star Schema

Check Your Progress 2:

1:

Star Schema	Snowflake Schema
It is a logical arrangement of one fact table surrounded by other dimension tables like a star.	It is a logical arrangement of one fact table with dimension tables and further dimension tables are normalized to other dimensions
It requires a single join SQL command to fetch the data	It requires many joins SQL command to fetch the data
Simple Database design and respond to query time is very less	Complex database design and respond time to queries is high
The data is not normalized. High level of redundancy	The data is normalized so low level of redundancy.

2: a. Star Schema of Hospital Management

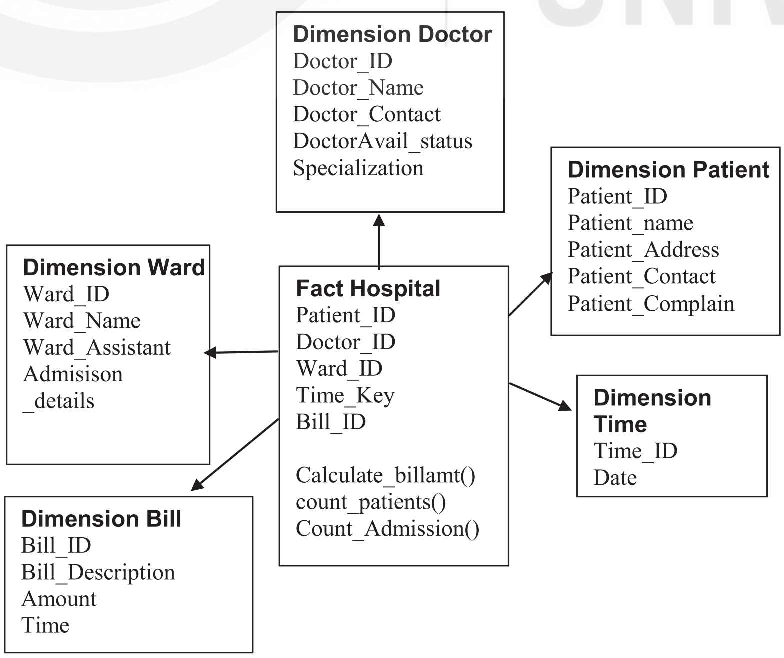

Figure 9 : Fact Schema of Hospital Management System

b. Snowflake Schema of Hospital Management

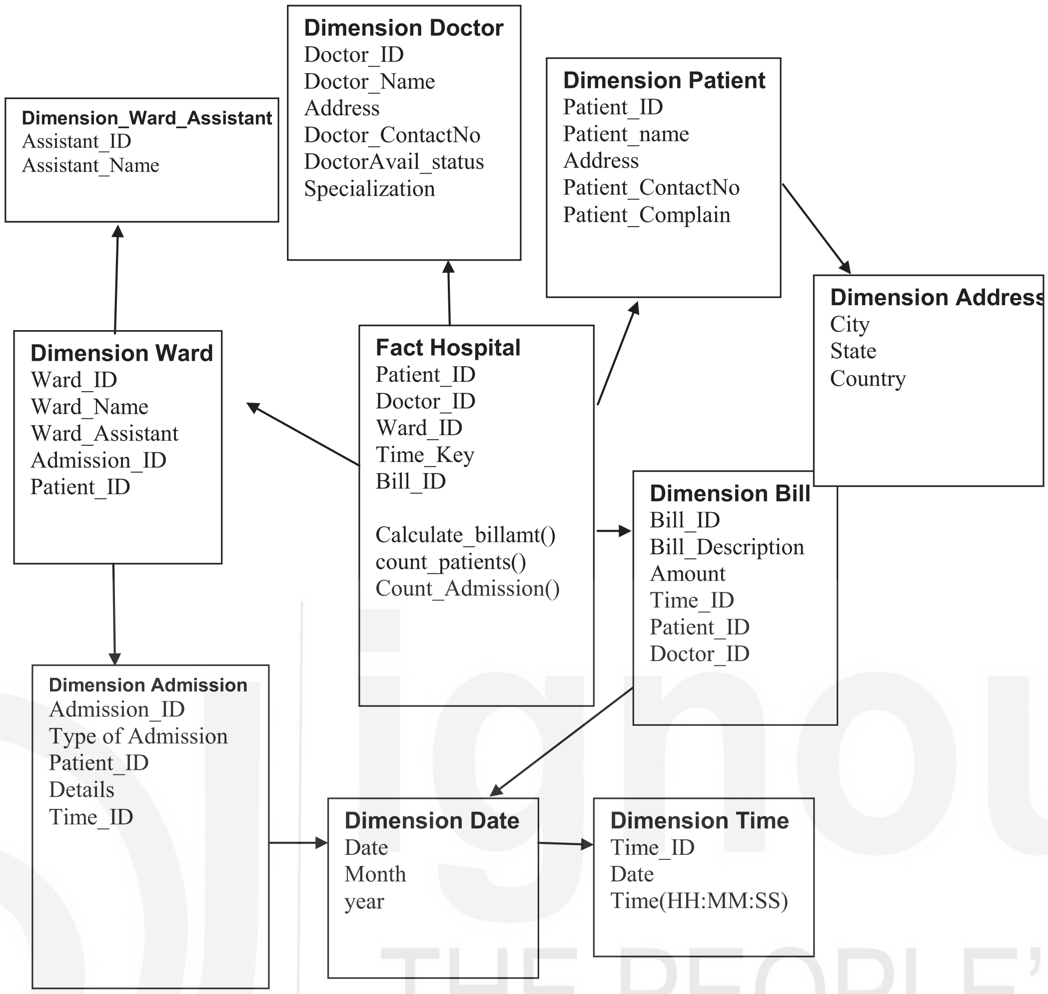

Figure 10: Snowflake Schema of Hospital Management System

☞ **Check Your Progress 3:**

1.

Limitations of Aggregate fact tables: Aggregate tables take lot of time to scan the rows of the base fact table. So, there will be more tables to manage. The size of aggregates in computing can be costly. Based on the greedy approach the size of aggregates is decided using hashing technique. If there are n dimensions in the table, then there can be 2n possible aggregates. The load on the data warehouse becomes more complex.

3.14 FURTHER READINGS

- Building the Data Warehouse, William H. Inmon, Wiley, 4th Edition, 2005.
- Data Warehousing Fundamentals, Paulraj Ponnaiah, Wiley Student Edition
- Data Warehousing, Reema Thareja, Oxford University Press.
- Data Warehousing, Data Mining & OLAP, Alex Berson and Stephen J.Smith, Tata McGraw – Hill Edition, 2016.